

HYPOTHESIS TESTING

- ❖ Hypothesis testing is a statistical method used to make decisions or inferences about a population parameter based on sample data.
- ❖ It evaluates whether the observed data supports a given hypothesis or not.
- ❖ Two Types of Hypotheses:
 - ❖ Null Hypothesis (H_0): Assumes no effect or no difference (status quo).
 - ❖ Alternative Hypothesis (H_1): Represents the claim we are testing, such as the presence of an effect or a difference.

NULL HYPOTHESIS (H_0)

- ❖ The null hypothesis (H_0) represents the default or status quo assumption.
- ❖ It assumes that there is no effect, no difference, or no relationship between variables. It is the statement that we test against.
- ❖ Mathematical Representation:
 - ❖ $H_0: \mu_1 = \mu_2$: No difference in population means.
 - ❖ $H_0: p = 0.5$: The proportion is equal to 0.5.
- ❖ Examples:
 - ❖ In medical trials: The null hypothesis might state that a new drug has no effect compared to a placebo (i.e., the drug does not improve patient outcomes).
 - ❖ In manufacturing: The null hypothesis might state that the average weight of a product is exactly 500 grams.
 - ❖ **The null hypothesis is assumed to be true unless there is strong evidence to reject it.**

ALTERNATIVE HYPOTHESIS (H_1)

- ❖ The alternative hypothesis (H_1) represents the statement we are trying to find evidence for. It suggests that there is an effect, a difference, or a relationship between variables.
- ❖ Types of Alternative Hypotheses:
 - ❖ **Two-tailed test:** The alternative hypothesis is that there is a difference, but the direction of the difference is not specified.
 - ❖ $H_1: \mu_1 \neq \mu_2$
 - ❖ **One-tailed test:** The alternative hypothesis specifies the direction of the difference (either greater or less).
 - ❖ $H_1: \mu_1 > \mu_2$ **or** $H_1: \mu_1 < \mu_2$
 - ❖ In medical trials: The alternative hypothesis might state that the new drug improves patient outcomes compared to a placebo. (One-tailed test)
 - ❖ In manufacturing: The alternative hypothesis might state that the average weight of a product is not equal to 500 grams (it could be either higher or lower). (Two-tailed test)

SIGNIFICANCE LEVEL (ALPHA)

- ❖ The significance level (α) is the probability of rejecting the null hypothesis when it is actually true.
- ❖ It is the threshold for deciding whether the observed data is unusual under the null hypothesis.
- ❖ Common Values:
 - ❖ $\alpha=0.05$: The most commonly used significance level, meaning a 5% risk of rejecting H_0 when it is true.
 - ❖ $\alpha=0.01$: Stricter threshold, meaning a 1% risk.
- ❖ Interpretation:
 - ❖ If the p-value is less than α , we reject the null hypothesis.
 - ❖ If the p-value is greater than α , we fail to reject the null hypothesis.

WHAT IS A P-VALUE?

- ❖ The p-value is the probability of observing the test statistic (or more extreme results) assuming that the null hypothesis (H_0) is true.
- ❖ It quantifies the evidence against the null hypothesis.
- ❖ Range:
 - ❖ The p-value ranges from 0 to 1.
 - ❖ Smaller p-values indicate stronger evidence against H_0 .

HOW TO INTERPRET THE P-VALUE?

- ❖ Compare the p-value to the Significance Level (α):
 - ❖ If $p\text{-value} < \alpha$: Reject the null hypothesis (H_0).
 - ❖ If $p\text{-value} \geq \alpha$: Fail to reject the null hypothesis.
- ❖ Interpretation Example:
 - ❖ If $\alpha=0.05$ and the p-value is 0.02, there is a 2% chance of observing the test result (or more extreme) if the null hypothesis is true. Since 0.02 is less than 0.05, we reject H_0 .

HOW TO INTERPRET THE P-VALUE?

- ❖ Scenario: Testing whether a new medication lowers blood pressure compared to a placebo.
- ❖ Null Hypothesis H_0 : The medication has no effect (mean difference in blood pressure = 0).
- ❖ Result: After conducting a statistical test, the p-value is calculated to be 0.03.
- ❖ Interpretation:
- ❖ If $\alpha=0.05$, we reject the null hypothesis since 0.03 is less than 0.05.
- ❖ This suggests that there is strong evidence that the medication reduces blood pressure.

WE DO NOT ACCEPT THE NULL HYPOTHESIS

❖ Rejecting the Null Hypothesis (H_0):

- ❖ If the evidence from the sample data suggests that the null hypothesis is unlikely to be true (i.e., $p\text{-value} < \alpha$), we reject the null hypothesis.
- ❖ This means there is enough evidence to support the alternative hypothesis.

❖ Failing to Reject the Null Hypothesis:

- ❖ If the evidence is insufficient (i.e., $p\text{-value} \geq \alpha$), we fail to reject the null hypothesis.
- ❖ This does not mean that H_0 is true; it means we do not have enough evidence to conclude otherwise.

❖ Important Note:

- ❖ We never “accept” the null hypothesis; we either reject it or fail to reject it.

TYPE I AND TYPE II ERRORS

- ❖ In hypothesis testing, two types of errors can occur when making decisions:
 - ❖ Type I Error (α): Rejecting the null hypothesis when it is actually true.
 - ❖ Type II Error (β): Failing to reject the null hypothesis when it is actually false.
- ❖ Why are they important?:
 - ❖ These errors represent the risks we take in decision-making during hypothesis testing, affecting the reliability of conclusions.

TYPE I ERROR — FALSE POSITIVE

- ❖ A Type I Error occurs when we reject the null hypothesis (H_0) when it is actually true. This means we conclude there is an effect or difference when, in reality, there is none.
- ❖ Probability of Type I Error (α):
 - ❖ The significance level α (e.g., 0.05) represents the probability of committing a Type I Error.
 - ❖ Example: If $\alpha=0.05$, there is a 5% chance of rejecting a true null hypothesis.
- ❖ Real-World Example:
 - ❖ In medical testing, a Type I Error would be concluding that a new drug works when it actually has no effect.

TYPE II ERROR — FALSE NEGATIVE

- ❖ A Type II Error occurs when we fail to reject the null hypothesis (H_0) when it is actually false. This means we conclude that there is no effect or difference when, in reality, there is one.
- ❖ Probability of Type II Error (β):
 - ❖ The probability of committing a Type II Error is denoted by β , and it depends on the sample size, effect size, and significance level.
- ❖ Real-World Example:
 - ❖ In medical testing, a Type II Error would be failing to detect that a new drug works when it actually does.

TYPE I AND TYPE II ERRORS

❖ Trade-off Between Errors:

- ❖ Reducing the chance of a Type I Error (lowering α) increases the risk of a Type II Error (β).
- ❖ Increasing sample size can help reduce both errors by providing more reliable estimates of the true effect.

❖ Importance of Context:

- ❖ In critical scenarios (e.g., medical trials), the consequences of Type I Error (false positive) may be more severe, so a lower α (e.g., 0.01) is chosen.
- ❖ In other situations, Type II Error (false negative) may be more concerning, requiring a higher power ($1 - \beta$) to ensure the true effect is detected.

TYPE I AND TYPE II ERRORS

Scenario	Null Hypothesis (H_0)	Type I Error (False Positive)	Type II Error (False Negative)
Court of Law	The defendant is innocent.	Convicting an innocent person (False conviction).	Letting a guilty person go free (False acquittal).
Fire Alarm in a Building	There is no fire in the building.	The fire alarm goes off, but there is no fire (False alarm).	The fire alarm doesn't go off when there is a fire (Failure to detect fire).
Medical Test	The patient does not have the disease.	The test says the patient has the disease when they don't (False positive).	The test says the patient doesn't have the disease when they do (Missed diagnosis).
Product Quality Control	The batch of products meets quality standards.	Rejecting a batch that is actually good (Unnecessary rejection).	Accepting a batch with defects (Passing faulty products).
Drug Efficacy Trial	The new drug is not effective.	Concluding the drug works when it doesn't (Unnecessary approval).	Concluding the drug doesn't work when it does (Missed treatment opportunity).
Security Screening	The passenger is not a threat.	Flagging a safe passenger as a threat (False alarm).	Letting a dangerous passenger pass through (Failure to detect threat).

POWER OF THE TEST

- ❖ The power of a test is the probability of correctly rejecting the null hypothesis when the alternative hypothesis is true (i.e., avoiding a Type II Error).
- ❖ $\text{Power} = 1 - \beta$.
- ❖ Factors Influencing Power:
 - ❖ Sample Size: Larger samples increase power and reduce β .
 - ❖ Effect Size: Larger effects are easier to detect, increasing power.
 - ❖ Significance Level (α): A higher α increases power, but also increases the chance of a Type I Error.
- ❖ Target Power:
 - ❖ Typically, researchers aim for a power of 0.80 (80%), meaning there is an 80% chance of detecting a true effect.

STATISTICAL VS. PRACTICAL SIGNIFICANCE

❖ Statistical Significance:

- ❖ Refers to the likelihood that the observed difference or relationship in the data is not due to chance.
- ❖ Determined through hypothesis testing (e.g., p-values).
- ❖ If a result is statistically significant, it means the evidence suggests an effect exists in the population.

❖ Practical Significance:

- ❖ Refers to the real-world relevance or importance of the result.
- ❖ Even if a result is statistically significant, it may not be large enough to have practical consequences or justify changes.

STATISTICAL VS. PRACTICAL SIGNIFICANCE

Feature	Statistical Significance	Practical Significance
Definition	Probability that the result is not due to chance	Whether the result is meaningful or impactful in real life
Measurement	Determined by p-value and comparison to significance level α .	Determined by effect size and context
Focus	Focuses on whether an effect exists	Focuses on whether the effect is large enough to matter
Interpretation	A result may be statistically significant but not useful	A result should be large enough to justify a real-world action
Example	A p-value of 0.03 suggests a statistically significant result	A 0.1% improvement in sales may not justify the cost of change

SAMPLE SIZE CALCULATION

- ❖ The sample size is the number of observations needed to achieve a specific level of precision in a hypothesis test or estimate.
- ❖ The margin of error (E) is the range within which the true population parameter is expected to lie, given the sample estimate.
- ❖ Importance:
 - ❖ A larger sample size reduces the margin of error, making estimates more precise.
 - ❖ A smaller sample size leads to larger margins of error and less precise estimates.

SAMPLE SIZE CALCULATION

❖ Margin of Error (E):

- ❖ Represents the maximum expected difference between the true population parameter and the sample estimate.
- ❖ Smaller margins of error require larger sample sizes.

❖ Confidence Level:

- ❖ The probability that the margin of error contains the true population parameter (e.g., 95% confidence level).
- ❖ A higher confidence level requires a larger sample size to maintain the same margin of error.

SAMPLE SIZE CALCULATION

- ❖ Scenario: You want to estimate the average time students spend studying per day. You aim for a 95% confidence level and a margin of error of 1 hour.
- ❖ Explanation:
 - ❖ A 95% confidence level means that you are 95% confident that the true average study time for all students falls within your estimated range.
 - ❖ If the sample mean is 4 hours with a margin of error of 1 hour, the confidence interval would be:
 - ❖ $4 \text{ hours} \pm 1 \text{ hour} = [3 \text{ hours}, 5 \text{ hours}]$
 - ❖ This means you are 95% confident that the true average study time for all students is between 3 and 5 hours per day.
 - ❖ If you want to reduce the margin of error to 0.5 hours, the required sample size would need to increase to ensure the same level of precision.

SAMPLE SIZE FORMULA FOR MEANS

❖ Formula:

$$❖ n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{E} \right)^2$$

❖ $Z_{\alpha/2}$: Z-value corresponding to the confidence level (e.g., 1.96 for 95% confidence).

❖ σ : Population standard deviation (assumed to be known).

❖ E: Desired margin of error.

❖ The formula shows that the sample size increases as the margin of error (E) decreases, or as the confidence level increases (larger $Z_{\alpha/2}$).

SAMPLE SIZE FORMULA FOR MEANS

- ❖ Problem: You want to estimate the average time students spend studying, with a margin of error of 1 hour, at a 95% confidence level. The population standard deviation (σ) is known to be 4 hours.
- ❖ Solution:
- ❖ Use $Z_{\alpha/2} = 1.96$, $\sigma = 4$, and $E = 1$:
- ❖
$$n = \left(\frac{1.96 \times 4}{1} \right)^2 = (7.84)^2 = 61.46 \approx 62$$
- ❖ Result: A sample size of 62 is needed to achieve a margin of error of 1 hour with 95% confidence.

SAMPLE SIZE FORMULA FOR PROPORTIONS

❖ Formula:

$$❖ n = \left(\frac{Z_{\alpha/2} \cdot \sqrt{p(1-p)}}{E} \right)^2$$

❖ p : Estimated proportion of the population.

❖ E : Desired margin of error.

❖ $Z_{\alpha/2}$: Z-value corresponding to the confidence level (e.g., 1.96 for 95%).

❖ This formula calculates the sample size needed to estimate a population proportion (e.g., the percentage of people who favor a candidate) with a specific margin of error and confidence level.

SAMPLE SIZE FORMULA FOR PROPORTIONS

❖ What if p is Not Known?

$$❖ n = \left(\frac{Z_{\alpha/2} \cdot \sqrt{p(1-p)}}{E} \right)^2$$

❖ p : Estimated proportion of the population.

❖ E : Desired margin of error.

❖ $Z_{\alpha/2}$: Z-value corresponding to the confidence level (e.g., 1.96 for 95%).

❖ This formula calculates the sample size needed to estimate a population proportion (e.g., the percentage of people who favor a candidate) with a specific margin of error and confidence level.

SAMPLE SIZE FORMULA FOR PROPORTIONS

❖ Problem: You want to estimate the proportion of students who own a laptop with a margin of error of 5%, at a 95% confidence level. You estimate the proportion to be 50% ($p=0.5$).

❖ Solution:

❖ Use $Z_{\alpha/2}=1.96$, $p=0.5$, and $E=0.05$:

$$\text{❖ } n = \left(\frac{1.96 \times \sqrt{0.5(1-0.5)}}{0.05} \right)^2 = \left(\frac{0.98}{0.05} \right)^2 = 384.16 \approx 385$$

❖ Result: A sample size of 385 is needed to estimate the proportion of students who own a laptop with a margin of error of 5% and 95% confidence.

POINT AND INTERVAL ESTIMATES

- ❖ Point Estimate:
- ❖ A point estimate provides a single value as an estimate of a population parameter (e.g., the sample mean $\bar{x}$ is a point estimate of the population mean μ).
- ❖ Interval Estimate:
- ❖ An interval estimate provides a range of values within which the true population parameter is expected to lie with a certain level of confidence.
- ❖ Common interval estimates include **confidence intervals** and **prediction intervals**.

POINT AND INTERVAL ESTIMATES

- ❖ Confidence Interval (CI):
- ❖ A confidence interval gives a range within which the true **population parameter** (e.g., population mean μ) is likely to lie with a certain probability (confidence level).
- ❖ Formula for a Confidence Interval (for mean with known standard deviation):
- ❖ $CI = \bar{X} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$
- ❖ Where $\bar{X}$ is the sample mean, $Z_{\alpha/2}$ is the Z-value for the confidence level, σ is the population standard deviation, and n is the sample size.

POINT AND INTERVAL ESTIMATES

- ❖ Prediction Interval (PI):
- ❖ A prediction interval gives a range within which a **single new observation** is likely to fall.
- ❖ Formula for a Prediction Interval:
- ❖ $PI = \bar{X} \pm Z_{\alpha/2} \cdot \sqrt{\sigma^2 \cdot \left(1 + \frac{1}{n}\right)}$
- ❖ The prediction interval is wider than the confidence interval because it accounts for both the uncertainty in the sample mean and the variability of individual observations.

POINT AND INTERVAL ESTIMATES

- ❖ You want to estimate the average test score for a class and predict the score for an individual student. The sample mean test score is 70 with a standard deviation of 10 from a sample size of 25, and you want a 95% confidence level.
- ❖ Confidence Interval: $CI = \bar{X} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$
- ❖ $CI = 70 \pm 1.96 \cdot \frac{10}{\sqrt{25}} = 70 \pm 1.96 \cdot 2 = 70 \pm 3.92 = [66.08, 73.92]$
- ❖ You are 95% confident that the true average test score lies between 66.08 and 73.92.
- ❖ Prediction Interval: $PI = \bar{X} \pm Z_{\alpha/2} \cdot \sqrt{\sigma^2 \cdot \left(1 + \frac{1}{n}\right)}$
- ❖ $PI = \bar{X} \pm Z_{\alpha/2} \cdot \sqrt{\sigma^2 \cdot \left(1 + \frac{1}{n}\right)} = 70 \pm 1.96 \cdot \sqrt{10^2 \cdot \left(1 + \frac{1}{25}\right)} = [50.01, 89.99]$
- ❖ Interpretation: You are 95% confident that a single new student's score will fall between 50.01 and 89.99.

TESTS FOR MEANS, VARIANCES, AND PROPORTIONS

- ❖ Hypothesis testing is a statistical method used to decide whether to reject or fail to reject a null hypothesis (H_0), based on sample data.
- ❖ It applies to various population parameters like means, variances, and proportions.
- ❖ Purpose: To determine if observed data provide enough evidence to reject a null hypothesis in favour of an alternative hypothesis H_A .
- ❖ Common hypothesis tests include tests for means (e.g., comparing averages), variances (e.g., comparing variability), and proportions (e.g., comparing percentages).

HYPOTHESIS TESTING PROCESS - STEPS

❖ 1. State the Hypotheses:

- ❖ Null Hypothesis (H_0): Assumes no effect or no difference.
- ❖ Alternative Hypothesis (H_1): Assumes an effect or a difference exists.

❖ 2. Select the Significance Level (α): Typically $\alpha=0.05$, representing a 5% chance of rejecting H_0 when it is true.

❖ 3. Calculate the Test Statistic: The test statistic measures how far the sample data are from the null hypothesis.

- ❖ The formula depends on the type of test (e.g., z-test, t-test, chi-square test).

❖ 4. Find the Critical Value: The critical value corresponds to the significance level and determines the decision rule for rejecting H_0 .

❖ 5. Make a Decision: Compare the test statistic with the critical value. If the test statistic exceeds the critical value (in absolute terms), reject H_0 .

- ❖ Alternatively, compare the p-value with α (if using software). If $p\text{-value} < \alpha$, reject H_0 .

HYPOTHESIS TESTING PROCESS - STEPS

❖ Test Statistic:

- ❖ A standardized value calculated from sample data that allows you to compare the sample against the null hypothesis.
- ❖ Examples:
 - ❖ Z-statistic for large sample sizes or known variance.
 - ❖ T-statistic for small sample sizes or unknown variance.
 - ❖ Chi-square statistic for tests on variances.

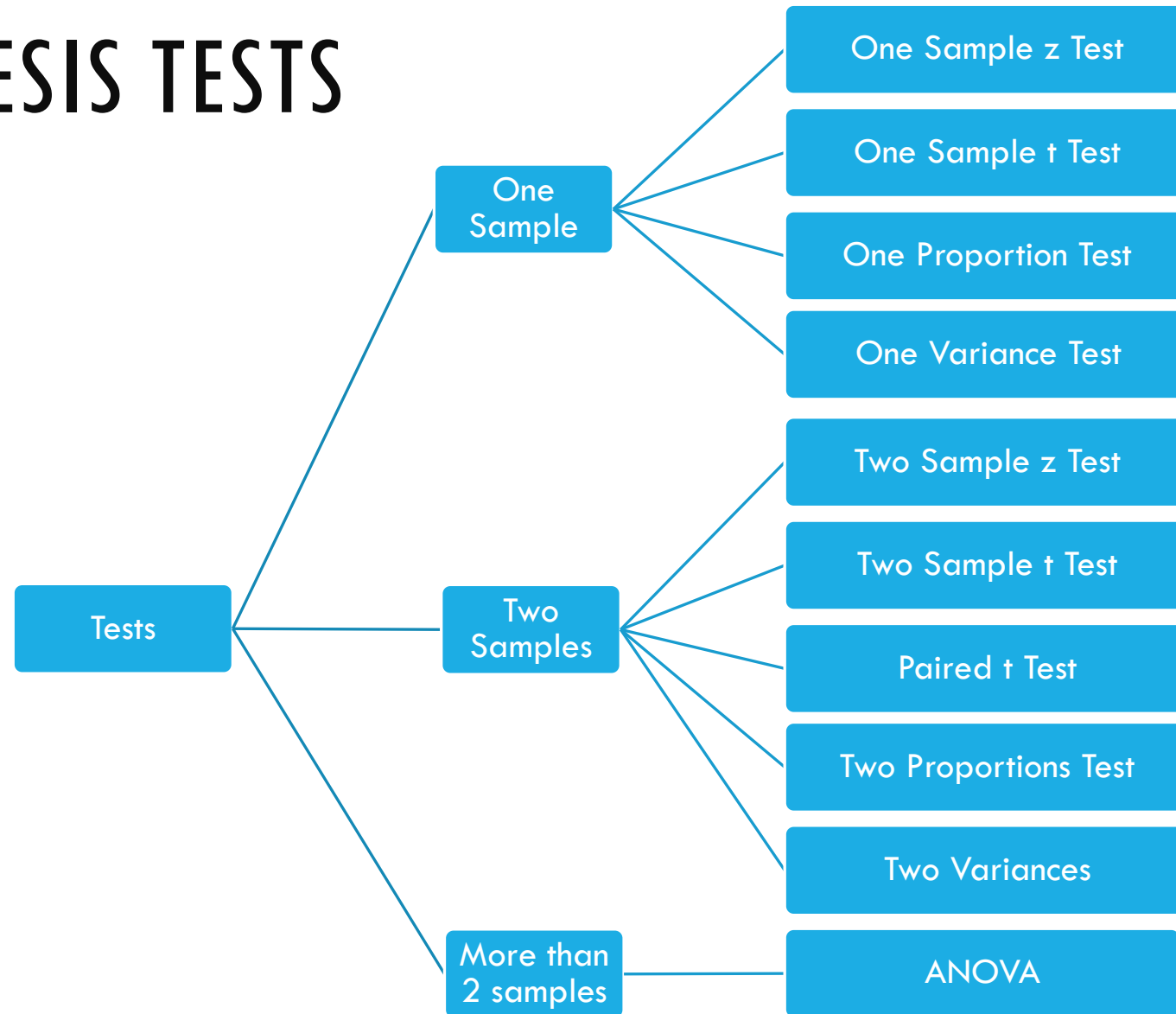
❖ Critical Value:

- ❖ The cutoff point in a distribution that determines whether to reject the null hypothesis.
- ❖ It is based on the significance level (α) and the type of test.
- ❖ Example:
 - ❖ For a 95% confidence level and a two-tailed test, the critical value for the Z-distribution is ± 1.96 .

HYPOTHESIS TESTING PROCESS - STEPS

- ❖ P-Value:
- ❖ The p-value represents the probability of observing the test results, or more extreme results, under the assumption that H_0 is true.
- ❖ If the p-value is less than α , we reject the null hypothesis.
- ❖ Interpretation:
 - ❖ p-value $< \alpha$: Reject H_0 (there is sufficient evidence to support H_A).
 - ❖ p-value $\geq \alpha$: Fail to reject H_0 (there is insufficient evidence to support H_A).
- ❖ Most statistical software provides p-values as the primary output for hypothesis tests, eliminating the need to manually calculate the test statistic and critical value.

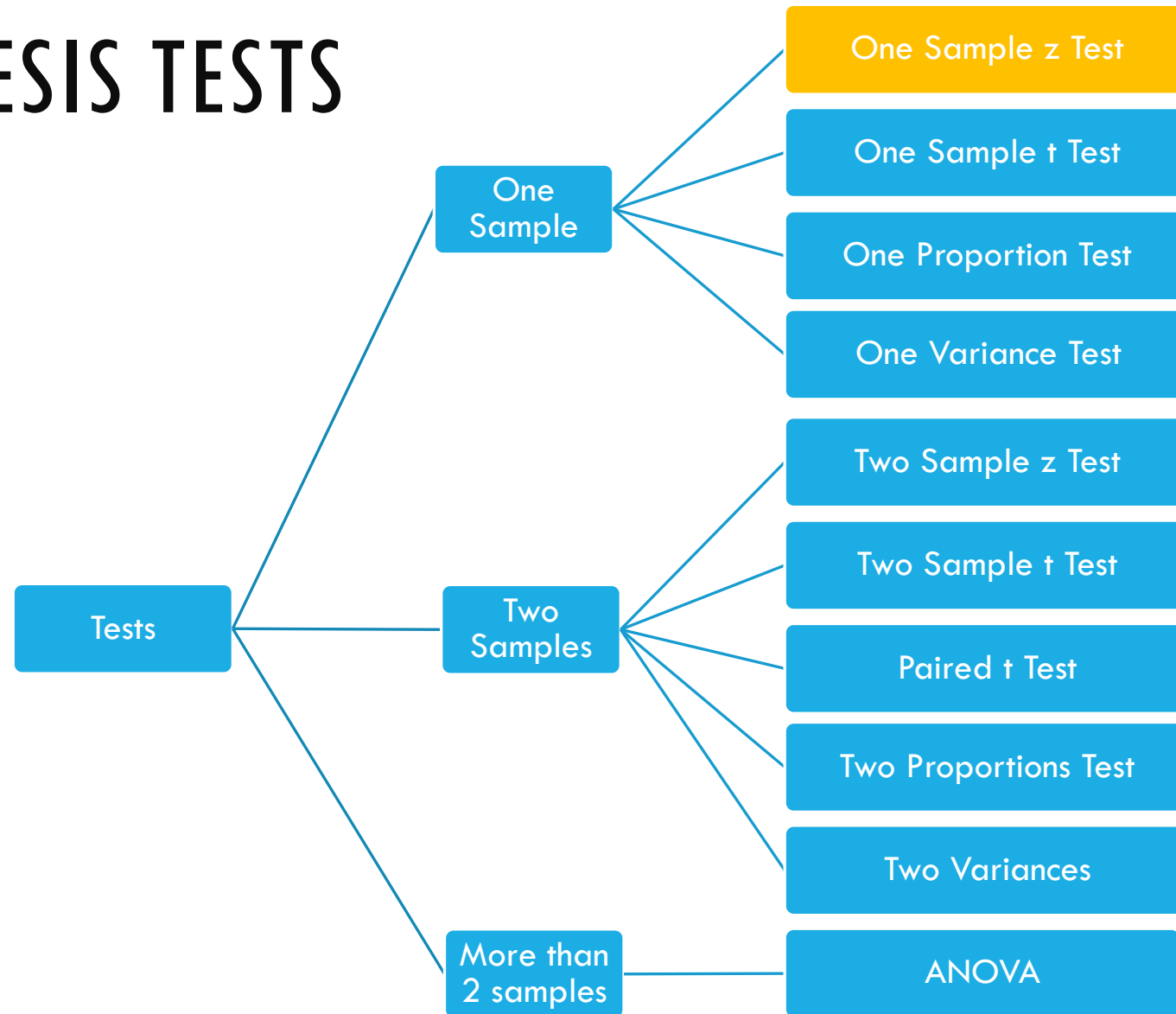
HYPOTHESIS TESTS



HYPOTHESIS TESTS

Test	When to Use	Hypothesis	Conditions
One Sample z Test	Testing the mean of one sample against a known population mean	$H_0: \mu = \mu_0$	Population variance σ^2 known, OR $n > 30$
One Sample t Test	Testing the mean of one sample when population variance is unknown	$H_0: \mu = \mu_0$	Population variance unknown, AND $n < 30$
One Proportion Test	Testing if a sample proportion matches a known population proportion	$H_0: p = p_0$	Sample proportion $\hat{p}$ vs known p_0
One Variance Test	Testing if a sample variance differs from a known population variance	$H_0: \sigma^2 = \sigma_0^2$	Uses Chi-square distribution
Two Sample z Test	Testing the difference between two sample means when variances are known	$H_0: \mu_1 = \mu_2$	Population variances known, OR $n > 30$
Two Sample t Test	Testing the difference between two sample means when variances are unknown	$H_0: \mu_1 = \mu_2$	Population variances unknown, AND small samples
Paired t Test	Comparing means of two related groups (e.g., before and after treatment)	$H_0: \mu_d = 0$	Paired data (dependent samples)
Two Proportions Test	Testing the difference between two sample proportions	$H_0: p_1 = p_2$	Comparing two proportions
Two Variances Test	Testing the equality of two population variances	$H_0: \sigma_1^2 = \sigma_2^2$	Uses F-distribution
ANOVA (Analysis of Variance)	Comparing means of more than two groups	$H_0: \mu_1 = \mu_2 = \mu_3 = \dots$	More than two independent groups

HYPOTHESIS TESTS



ONE-SAMPLE Z TEST

- ❖ A One-Sample Z-Test is used to determine whether the mean of a single sample significantly differs from a known or hypothesized population mean.
- ❖ It is typically used when the population standard deviation (σ) is known, OR the sample size is large ($n \geq 30$).

ONE-SAMPLE Z TEST - ASSUMPTIONS

- ❖ Population is Normally Distributed:

- ❖ The population should be normally distributed. If the sample size is large ($n \geq 30$), the Central Limit Theorem applies, making the test robust to deviations from normality.

- ❖ Known Population Standard Deviation (σ):

- ❖ The population standard deviation must be known, which is critical for calculating the z-statistic.

- ❖ Random Sampling:

- ❖ The sample must be randomly selected from the population, and each observation should be independent of others.

- ❖ Scale of Measurement:

- ❖ The data must be measured on an interval or ratio scale, allowing for meaningful arithmetic operations.

ONE-SAMPLE Z TEST - STEPS

- ❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): $\mu = \mu_0$ (The sample mean is equal to the population mean).
- ❖ Alternative Hypothesis (H_1): $\mu \neq \mu_0$ (two-tailed test), $\mu < \mu_0$ or $\mu > \mu_0$ (one-tailed tests).

- ❖ Choose the Significance Level (α):

- ❖ Typically set at 0.05 for a 95% confidence level.

- ❖ Calculate the Test Statistic (Z):

- ❖ Use the formula for the z-statistic (given in the next slide).

- ❖ Determine the Critical Value:

- ❖ Use the Z-table to find the critical value based on the chosen significance level.

- ❖ **Make a Decision:** Compare the calculated z-statistic with the critical z-value. If the absolute value of the z-statistic is greater than the critical value, reject the null hypothesis.

CALCULATING THE TEST STATISTIC (Z VALUE)

❖ Z-Statistic Formula:

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

❖ $\bar{X}$ = Sample mean.

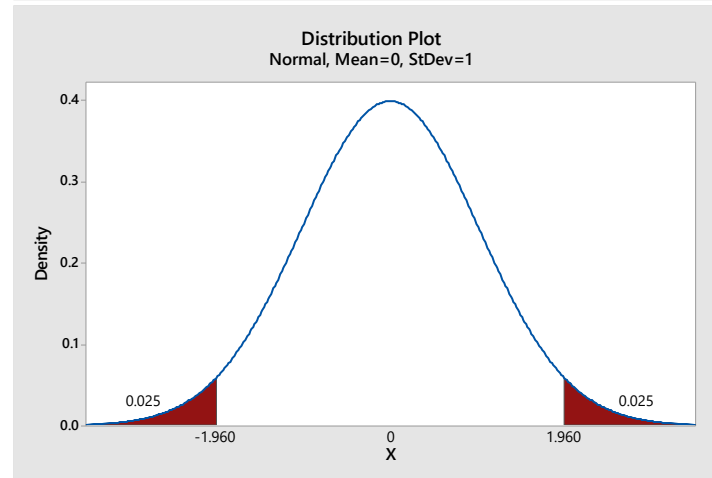
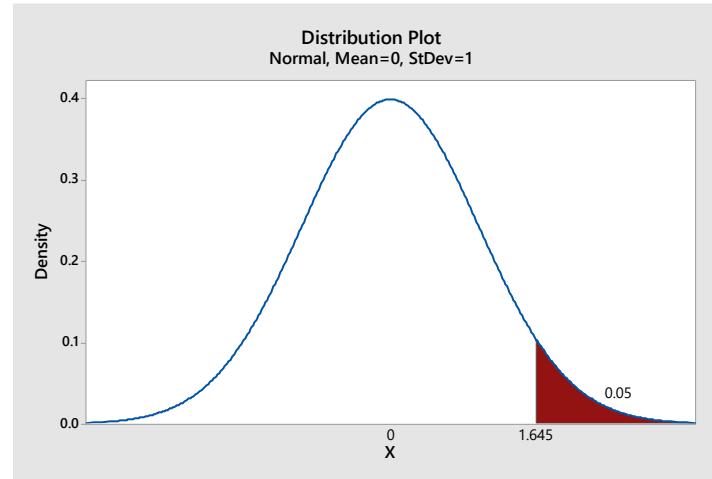
❖ μ_0 = Hypothesized population mean.

❖ σ = Population standard deviation.

❖ n = Sample size.

FINDING THE CRITICAL VALUE

z	0	1	2	3	4	5	6	7	8	9
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
3.7	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
3.8	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
3.9	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000



❖ $\alpha = 0.05$ One Tail

❖ Z Critical = 1.645

❖ $\alpha = 0.10$ One Tail

❖ Z Critical = 1.282

❖ $\alpha = 0.05$ Two Tails

❖ Z Critical = ± 1.96

❖ $\alpha = 0.10$ Two Tail

❖ Z Critical = ± 1.645

ONE-SAMPLE Z TEST (EXAMPLE - 1)

- ❖ A factory claims its perfume bottles have a mean volume of 150cc, with a population standard deviation of 2cc. A random sample of 100 bottles is selected, and the average volume is found to be 150.5cc. Test at a 95% confidence level **whether the mean volume differs from 150cc.**
- ❖ Step-by-Step Solution:
 - ❖ 1. Hypotheses:
 - ❖ $H_0: \mu = 150$
 - ❖ $H_1: \mu \neq 150$
 - ❖ 2. Significance Level:
 - ❖ $\alpha = 0.05$ for a 95% confidence level.

ONE-SAMPLE Z TEST (EXAMPLE)

❖ 3. Z-Statistic (or $Z_{\text{calculated}}$):

$$\text{❖ } Z = \frac{150.5 - 150}{\frac{2}{\sqrt{100}}} = \frac{0.5}{0.2} = 2.5$$

❖ 4. Critical Value:

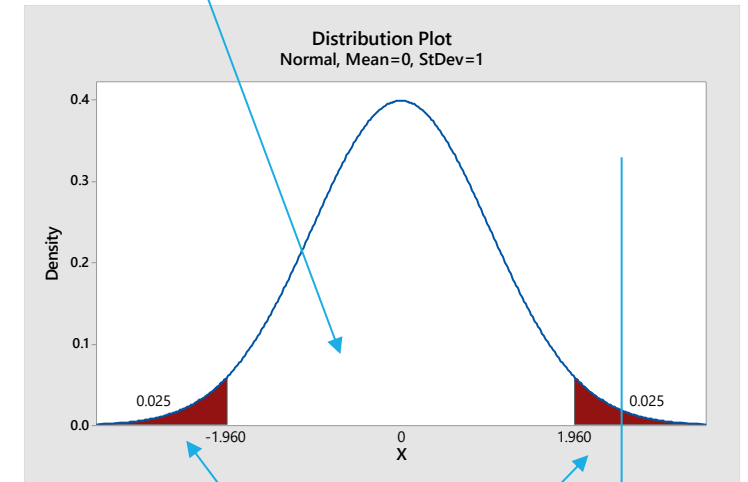
❖ For a two-tailed test with $\alpha=0.05$, the critical z-value is ± 1.96 .

❖ 5. Decision:

❖ Since $|Z|=2.5$ is greater than 1.96, we reject the null hypothesis.

❖ Conclusion: The data provide enough evidence to suggest the mean volume is different from 150cc.

Fail to Reject H_0



Reject H_0

ONE-SAMPLE Z TEST (EXAMPLE - 2)

- ❖ A factory claims its perfume bottles have a mean volume of 150cc, with a population standard deviation of 2cc. A random sample of 100 bottles is selected, and the average volume is found to be 150.5cc. Test at a 95% confidence level **whether the mean volume has increased** from 150cc.
- ❖ Step-by-Step Solution:
 - ❖ 1. Hypotheses:
 - ❖ $H_0: \mu \leq 150$ (The mean volume is 150cc).
 - ❖ $H_1: \mu > 150$ (The mean volume has increased from 150cc).
 - ❖ 2. Significance Level:
 - ❖ $\alpha = 0.05$ for a one-tailed test (since we're only testing for an increase).

ONE-SAMPLE Z TEST (EXAMPLE - 2)

❖ 3. Z-Statistic Calculation:

$$❖ Z = \frac{150.5 - 150}{\frac{2}{\sqrt{100}}} = \frac{0.5}{0.2} = 2.5$$

❖ 4. Critical Z-Value (for one-tailed test):

❖ From the Z-table, the critical value for $\alpha=0.05$ (one-tailed) is $Z=1.645$.

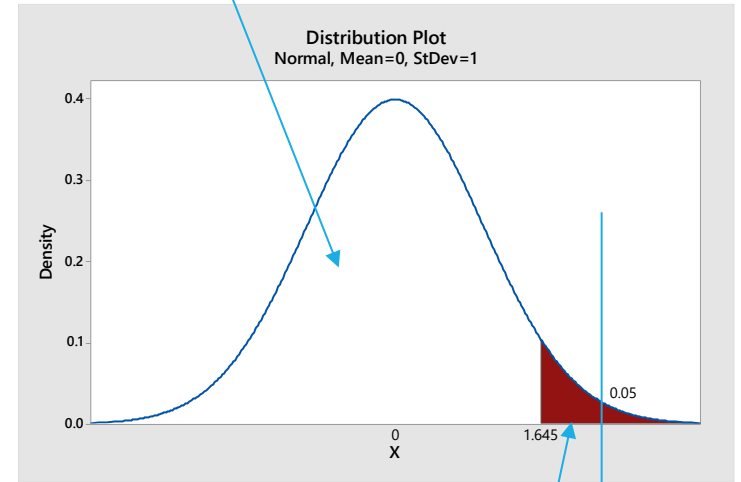
❖ 5. Decision:

❖ Since the calculated $Z=2.5$ is greater than the critical value $Z=1.645$, we reject the null hypothesis.

❖ Conclusion:

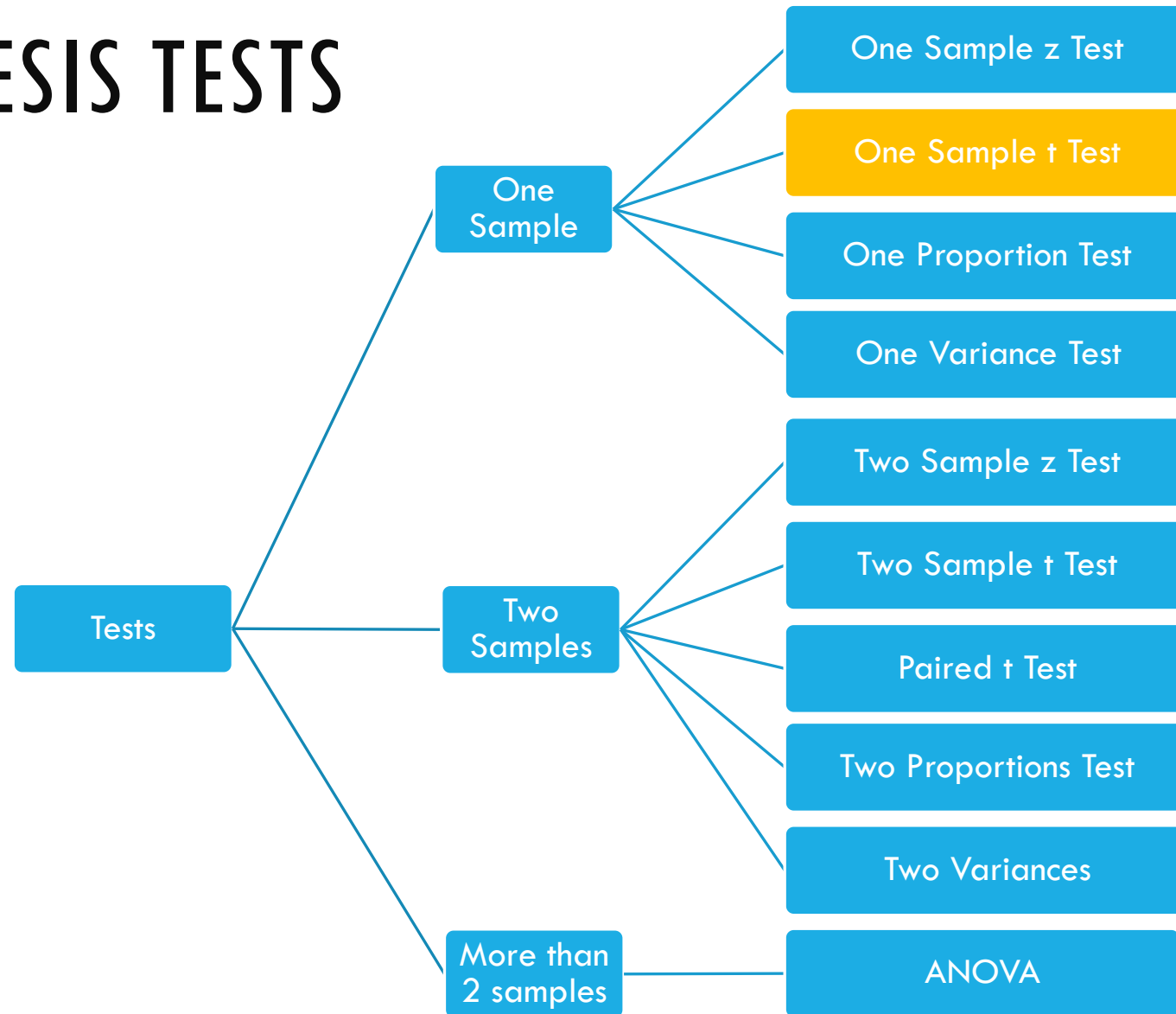
❖ There is enough statistical evidence to suggest that the mean volume of the perfume bottles has increased from 150cc at a 95% confidence level.

Fail to Reject H_0



Reject H_0

HYPOTHESIS TESTS



ONE-SAMPLE T TEST

- ❖ A One-Sample t-Test is used to determine whether the mean of a single sample significantly differs from a known or hypothesized population mean when the population standard deviation (σ) is unknown.
- ❖ It is typically used when the sample size is small ($n < 30$).

ONE-SAMPLE T TEST - ASSUMPTIONS

- ❖ Population is Normally Distributed:

- ❖ The population from which the sample is drawn should be normally distributed, especially for small samples ($n < 30$).

- ❖ Unknown Population Standard Deviation (σ):

- ❖ The population standard deviation is unknown, and we rely on the sample standard deviation (s) to estimate it.

- ❖ Random Sampling:

- ❖ The sample must be randomly selected from the population, and each observation should be independent of others.

- ❖ Scale of Measurement:

- ❖ The data must be measured on an interval or ratio scale, allowing for meaningful arithmetic operations.

ONE-SAMPLE T TEST - STEPS

❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): $\mu = \mu_0$ (The sample mean equals the population mean).
- ❖ Alternative Hypothesis (H_1): $\mu \neq \mu_0$ (two-tailed test), $\mu < \mu_0$ or $\mu > \mu_0$ (one-tailed tests).

❖ Choose the Significance Level (α):

- ❖ Typically set at 0.05 for a 95% confidence level.

❖ Calculate the Test Statistic (t):

- ❖ Use the formula for the t-statistic (given in the next slide).

❖ Determine the Critical t-Value:

- ❖ Use the t-table to find the critical value based on the significance level and degrees of freedom ($df = n - 1$).
- ❖ **Conclusion:** Compare the calculated t-statistic with the critical t-value. If the absolute value of the t-statistic exceeds the critical value, reject the null hypothesis.

FORMULA FOR CALCULATING T-STATISTIC

❖ t-Statistic Formula:

$$t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

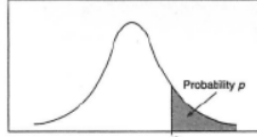
❖ $\bar{X}$ = Sample mean.

❖ μ_0 = Hypothesized population mean.

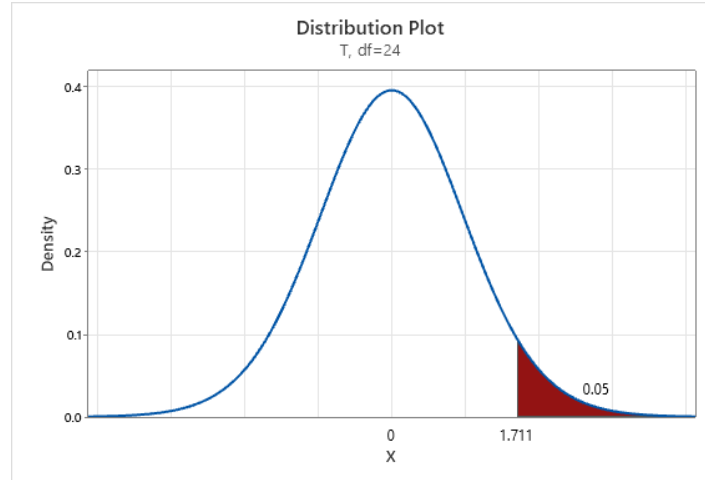
❖ s = Sample standard deviation.

❖ n = Sample size.

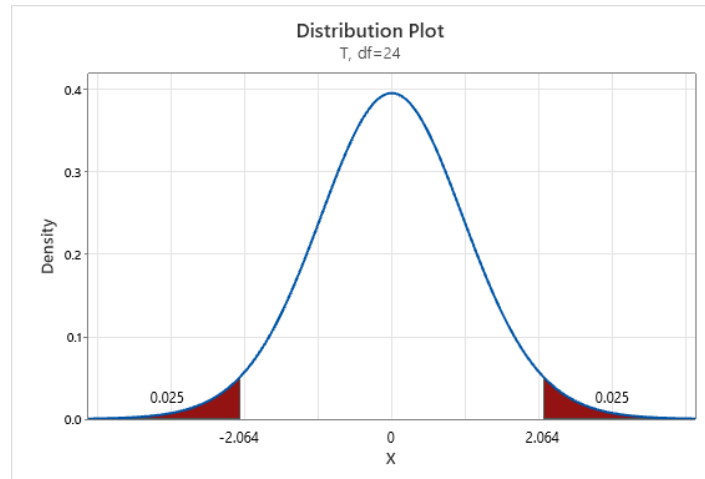
CRITICAL TEST STATISTIC (T VALUE)



df	TAIL PROBABILITY P											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646



- ❖ $\alpha = 0.05$ One Tails
- ❖ $Df = 24$
- ❖ $t \text{ Critical} = 1.711$



- ❖ $\alpha = 0.05$ Two Tails
- ❖ $Df = 24$
- ❖ $t \text{ Critical} = 2.064$

ONE-SAMPLE T TEST - EXAMPLE

- ❖ A factory claims its perfume bottles have a mean volume of 150cc, but the **population standard deviation is unknown**. A random sample of **25 bottles** is selected, and the average volume is found to be 151cc with a sample standard deviation of 1.5cc. Test at a 95% confidence level **whether the mean volume has increased** from 150cc.
- ❖ Step-by-Step Solution:
- ❖ 1. Hypotheses:
 - ❖ $H_0: \mu \leq 150$ (The mean volume is 150cc).
 - ❖ $H_1: \mu > 150$ (The mean volume has increased from 150cc).
- ❖ 2. Significance Level:
 - ❖ $\alpha = 0.05$ for a one-tailed test.

ONE-SAMPLE T TEST - EXAMPLE

❖ 3. t-Statistic Calculation:

$$❖ t = \frac{151-150}{\frac{1.5}{\sqrt{25}}} = \frac{1}{0.3} = 3.33$$

❖ 4. Degrees of Freedom:

$$❖ df = n - 1 = 25 - 1 = 24$$

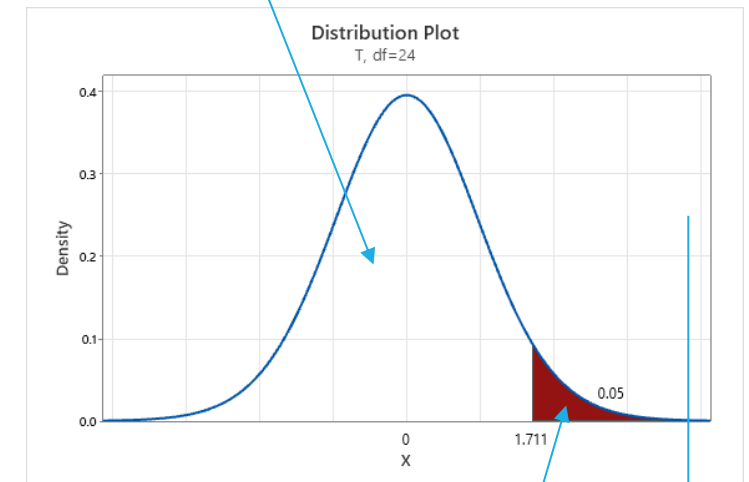
❖ Critical t-Value:

❖ From the t-table for $df=24$ and $\alpha=0.05$ (one-tailed), the critical t-value is 1.711.

❖ Decision:

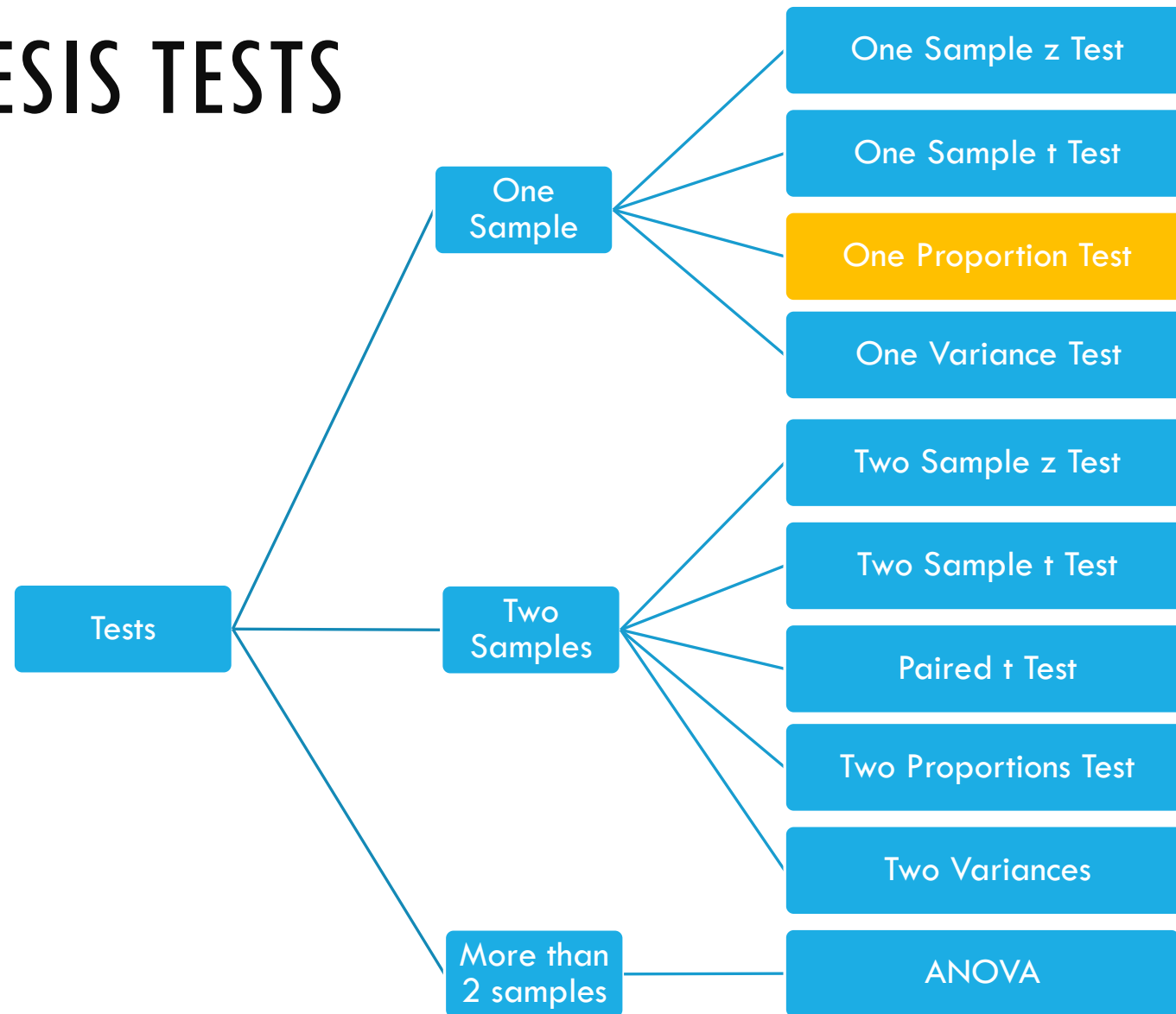
- ❖ Since the calculated $t=3.33$ is greater than the critical value $t=1.711$, we reject the null hypothesis.
- ❖ There is enough statistical evidence to suggest that the mean volume of the perfume bottles has increased from 150cc at a 95% confidence level.

Fail to Reject H_0



Reject H_0

HYPOTHESIS TESTS



ONE-PROPORTION TEST

- ❖ A One Proportion Test is used to determine whether the proportion of successes in a single sample differs significantly from a known or hypothesized population proportion.
- ❖ Purpose:
- ❖ It tests whether the sample proportion $\hat{p}$ differs from a population proportion p_0 .

ONE-PROPORTION TEST - ASSUMPTIONS

❖ Binary Data:

- ❖ The data should be categorical with two possible outcomes (e.g., success/failure, yes/no).

❖ Random Sampling:

- ❖ The sample must be randomly selected from the population, and each observation should be independent of others.

❖ Sample Size:

- ❖ The sample size must be large enough to use the normal approximation to the binomial distribution. This is generally satisfied if $n \cdot p_0 \geq 5$ and $n \cdot (1 - p_0) \geq 5$.
- ❖ This means that the data should have at least 5 successes and 5 failures. Some other references list this number as 10.

ONE-PROPORTION TEST - STEPS

❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): $p=p_0$ (The sample proportion is equal to the population proportion).
- ❖ Alternative Hypothesis (H_1): $p \neq p_0$ (two-tailed test), $p < p_0$ or $p > p_0$ (one-tailed tests).

❖ Choose the Significance Level (α):

- ❖ Typically set at 0.05 for a 95% confidence level.

❖ Calculate the Test Statistic (Z):

- ❖ Use the formula for the z-statistic (given in the next slide).

❖ Determine the Critical Z-Value:

- ❖ Use the Z-table to find the critical value based on the significance level and whether the test is one-tailed or two-tailed. (Similar to the One sample Z Test discussed earlier)
- ❖ **Make a Decision:** Compare the calculated z-statistic with the critical z-value. If the absolute value of the z-statistic is greater than the critical value, reject the null hypothesis.

ONE-PROPORTION TEST - EXAMPLE

$$z = \frac{p - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

- ❖ A factory claims that 80% of its products pass quality inspection. A random sample of 100 products is taken, and 75 products pass. Test at a 95% confidence level whether the proportion of products passing inspection is less than 80%.
- ❖ Step-by-Step Solution:
 - ❖ 1. Hypotheses:
 - ❖ $H_0: p \geq 0.80$ (The proportion of passing products is 80%).
 - ❖ $H_1: p < 0.80$ (The proportion of passing products is less than 80%).
 - ❖ 2. Significance Level:
 - ❖ $\alpha=0.05$ for a one-tailed test.

ONE-PROPORTION TEST - EXAMPLE

- ❖ Sample Proportion: $p=75/100=0.75$
- ❖ 3. Can we use the Normal Distribution approximation?
- ❖ $p_0 = 0.8, n=100$
- ❖ $np_0 = 100 \times 0.8 = 80$
- ❖ $n(1-p_0) = 100 \times 0.20 = 20$
- ❖ Both np_0 and $n(1-p_0)$ are greater than 5; hence we can use this formula.

$$z = \frac{p - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

ONE-PROPORTION TEST - EXAMPLE

❖ Sample Proportion: $p=75/100=0.75$

❖ 4. Z-Statistic Calculation:

$$❖ Z = \frac{0.75 - 0.80}{\sqrt{\frac{0.80(1-0.80)}{100}}} = -1.25$$

$$Z = \frac{p - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

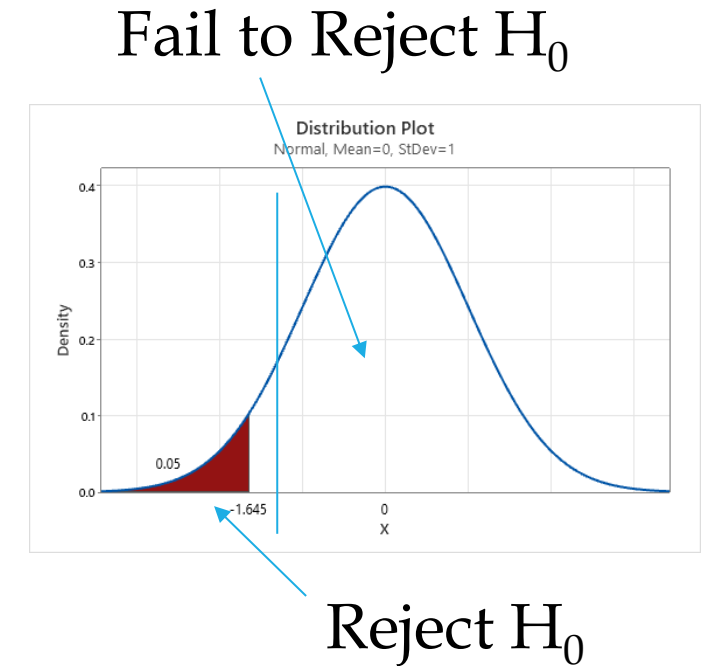
❖ 5. Critical Z-Value:

❖ From the Z-table for $\alpha=0.05$ (one-tailed), the critical z-value is -1.645.

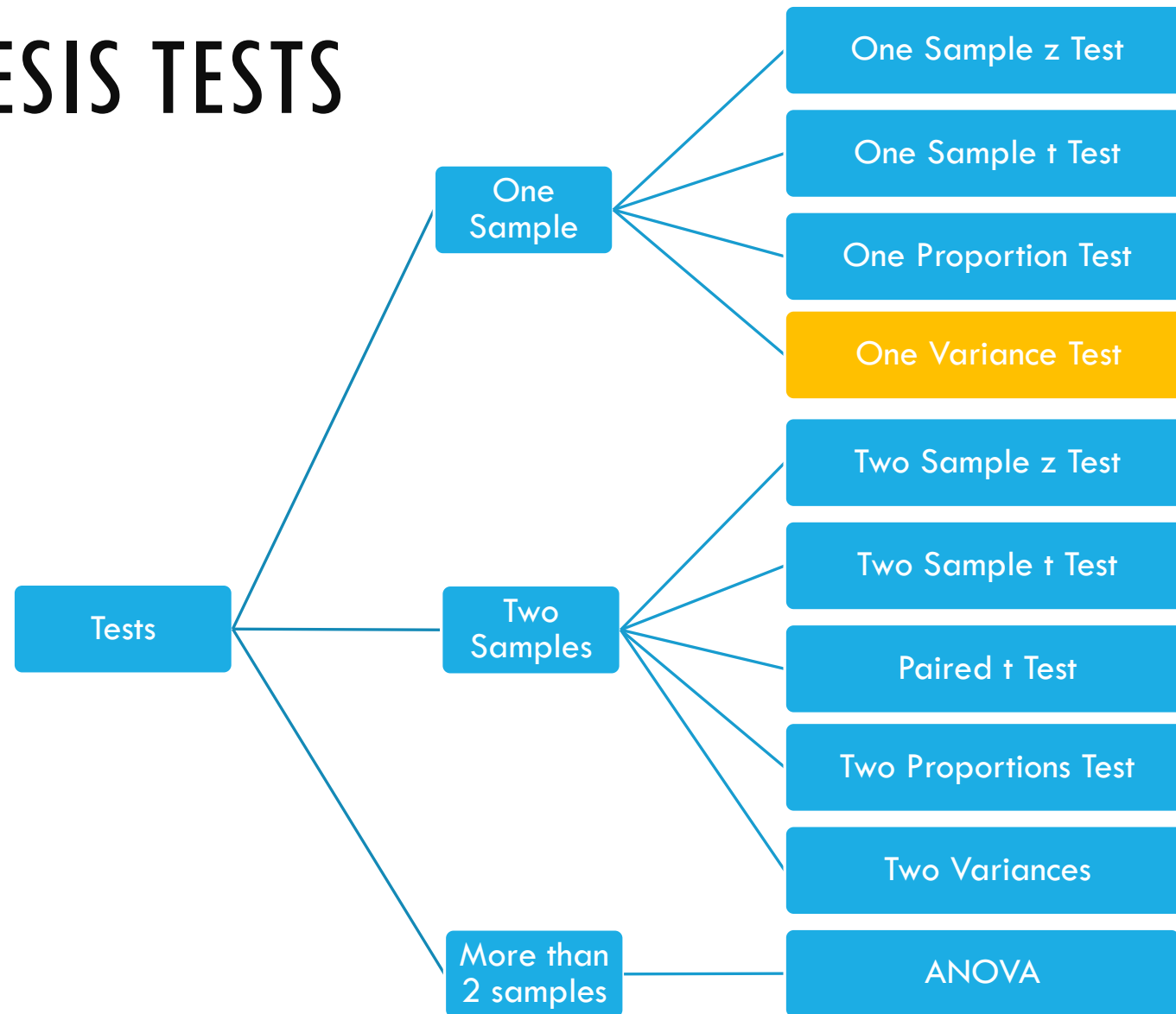
❖ 6. Decision:

❖ Since $Z=-1.25$ is greater than the critical value of -1.645 , we fail to reject the null hypothesis.

❖ There is not enough statistical evidence to suggest that the proportion of passing products is less than 80%.



HYPOTHESIS TESTS



ONE-VARIANCE TEST

- ❖ A one variance test is used to determine if the variance of a population differs from a specific value (i.e., whether the observed variance is significantly different from the hypothesized variance).
- ❖ Test Statistic:
- ❖ This test typically uses the Chi-Square test for variance.

SUMMARY OF VARIANCE TESTS

❖ Chi-Square Test:

- ❖ For testing the population variance against a specified value (One-variance test).
- ❖ For testing goodness-of-fit of a probability distribution to observed data.
- ❖ For testing independence between two categorical variables (Contingency Tables).

❖ F-Test:

- ❖ For testing equality of two variances from different populations.
- ❖ For testing equality of multiple means with the ANOVA technique (Analysis of Variance).

ONE-VARIANCE TEST - ASSUMPTIONS

❖ Population is Normally Distributed:

- ❖ The population from which the sample is drawn must follow a normal distribution.
- ❖ This assumption is important because the Chi-Square distribution is used, which assumes normality.

❖ Random Sampling:

- ❖ The sample must be randomly selected from the population, and each observation should be independent of others.

❖ Scale of Measurement:

- ❖ The data must be measured on an interval or ratio scale to calculate meaningful variance.

ONE-VARIANCE TEST — CHI-SQUARE TEST

❖ Chi-Square Test Statistic Formula:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

❖ χ^2 : Chi-square test statistic.

❖ n: Sample size.

❖ s^2 : Sample variance.

❖ σ_0^2 : Hypothesized population variance.

ONE-VARIANCE TEST — STEPS

- ❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): $s^2 = \sigma_0^2$ (The population variance is equal to the hypothesized variance).

- ❖ Alternative Hypothesis (H_1): $s^2 \neq \sigma_0^2$ (two-tailed), $s^2 < \sigma_0^2$ or $s^2 > \sigma_0^2$ (one-tailed).

- ❖ Set the Significance Level (α):

- ❖ Typically set at 0.05 for a 95% confidence level.

- ❖ Calculate the Test Statistic:

- ❖ Use the formula for the chi-square test statistic.

- ❖ Find the Critical Value:

- ❖ Use the chi-square distribution table with degrees of freedom ($df = n - 1$) to find the critical value.

Compare the calculated χ^2 value with the critical value. If the test statistic exceeds the critical value, reject H_0 .

ONE-VARIANCE TEST — EXAMPLE

- ❖ A manufacturer claims that the variance in the weight of their product is 4 grams². A sample of 10 products is tested, and the sample variance is found to be 5 grams². Test at a 90% confidence level whether the population **variance is different from 4 grams²**.
- ❖ Step-by-Step Solution:
- ❖ 1. Hypotheses:
 - ❖ $H_0: \sigma^2=4$ (The population variance is 4 grams²).
 - ❖ $H_1: \sigma^2 \neq 4$ (The population variance is different from 4 grams²).
- ❖ 2. Significance Level:
 - ❖ $\alpha=0.10$ (two-tailed test).

ONE-VARIANCE TEST — EXAMPLE

❖ 3. Degrees of Freedom:

❖ $df = n - 1 = 10 - 1 = 9$

❖ 4. Test Statistic:

❖ $\chi^2 = (n - 1) \left(\frac{s^2}{\sigma_0^2} \right) = (10 - 1) \times \frac{5}{4} = 9 \times \frac{5}{4} = 11.25$

❖ 5. Critical Values:

❖ From the Chi-Square Table, for $df=9$ and $\alpha=0.10$ (two-tailed):

❖ Lower critical value: $\chi^2_{(0.05,9)} = 3.325$

❖ Upper critical value: $\chi^2_{(0.95,9)} = 16.92$

❖ Decision:

❖ Since $3.325 < 11.25 < 16.92$, we fail to reject the null hypothesis.

❖ There is not enough evidence to suggest that the population variance differs from 4 grams².

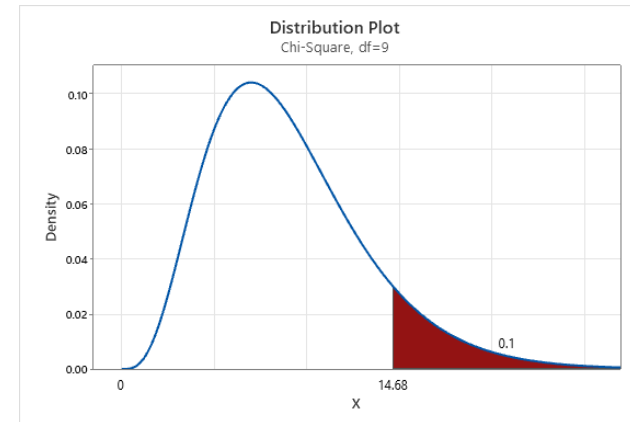
Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67

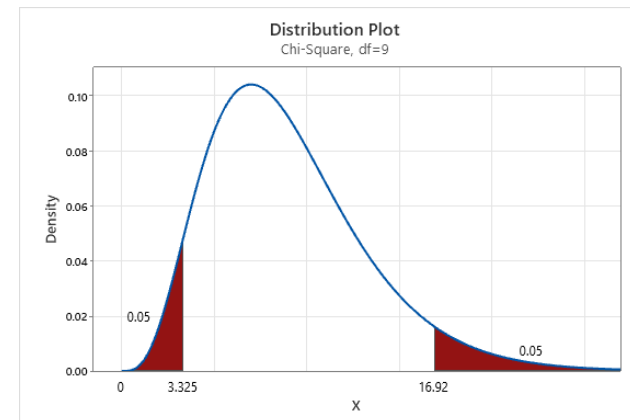
ONE-VARIANCE TEST — EXAMPLE

Percentage Points of the Chi-Square Distribution

Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48
8	1.647	2.733	3.490	5.071	7.344	10.22	13.36	15.51	20.09
9	2.088	3.325	4.168	5.899	8.343	11.39	14.68	16.92	21.67
10	2.558	3.940	4.865	6.737	9.342	12.55	15.99	18.31	23.21
11	3.053	4.575	5.578	7.584	10.341	13.70	17.28	19.68	24.72
12	3.571	5.226	6.304	8.438	11.340	14.85	18.55	21.03	26.22
13	4.107	5.892	7.042	9.299	12.340	15.98	19.81	22.36	27.69
14	4.660	6.571	7.790	10.165	13.339	17.12	21.06	23.68	29.14
15	5.229	7.261	8.547	11.037	14.339	18.25	22.31	25.00	30.58
16	5.812	7.962	9.312	11.912	15.338	19.37	23.54	26.30	32.00
17	6.408	8.672	10.085	12.792	16.338	20.49	24.77	27.59	33.41
18	7.015	9.390	10.865	13.675	17.338	21.60	25.99	28.87	34.80
19	7.633	10.117	11.651	14.562	18.338	22.72	27.20	30.14	36.19
20	8.260	10.851	12.443	15.452	19.337	23.83	28.41	31.41	37.57
22	9.542	12.338	14.041	17.240	21.337	26.04	30.81	33.92	40.29
24	10.856	13.848	15.659	19.037	23.337	28.24	33.20	36.42	42.98
26	12.198	15.379	17.292	20.843	25.336	30.43	35.56	38.89	45.64
28	13.565	16.928	18.939	22.657	27.336	32.62	37.92	41.34	48.28
30	14.953	18.493	20.599	24.478	29.336	34.80	40.26	43.77	50.89
40	22.164	26.509	29.051	33.660	39.335	45.62	51.80	55.76	63.69
50	27.707	34.764	37.689	42.942	49.335	56.33	63.17	67.50	76.15
60	37.485	43.188	46.459	52.294	59.335	66.98	74.40	79.08	88.38

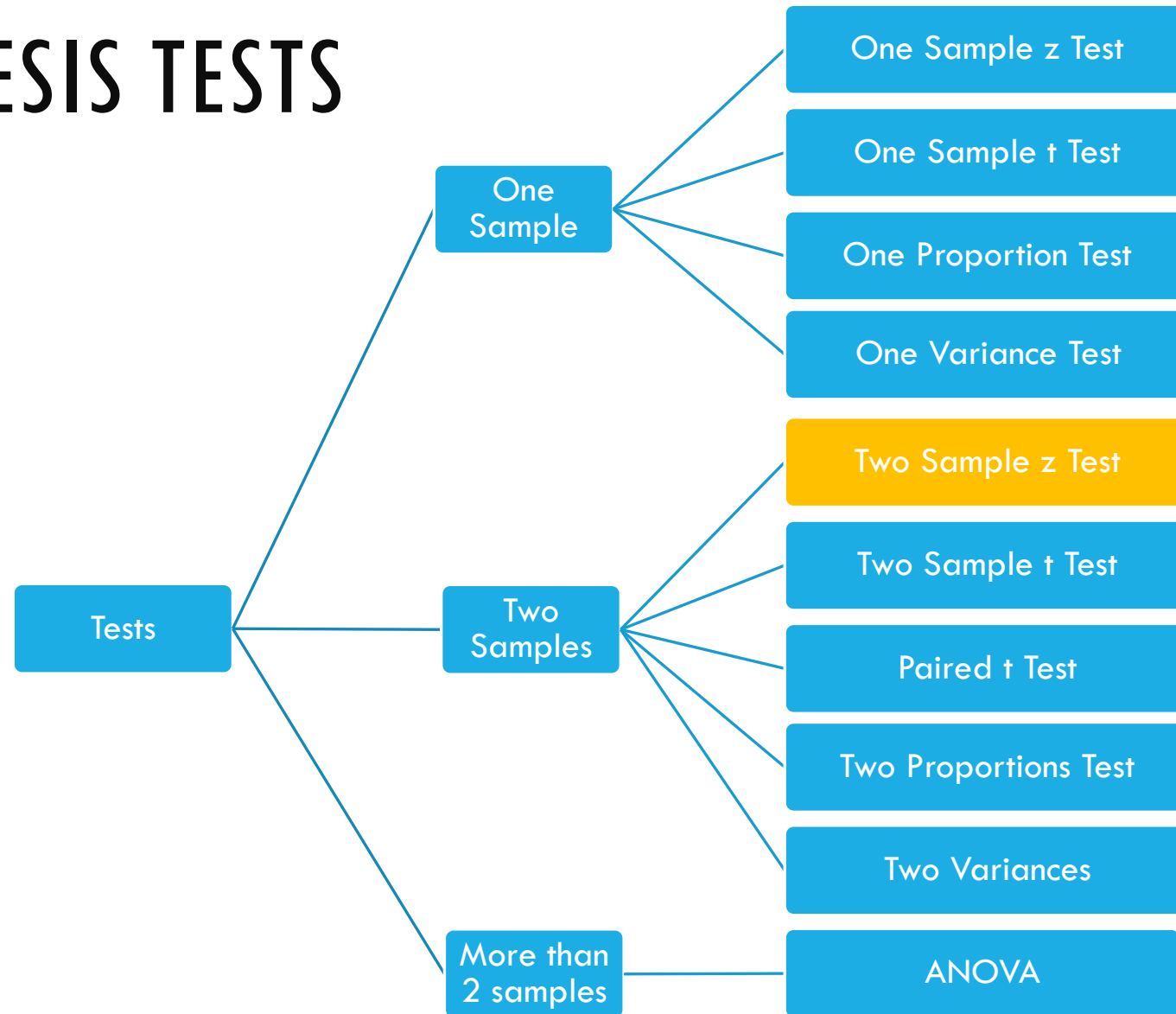


- ❖ $\alpha = 0.10$ One Tail
- ❖ $Df = 9$
- ❖ χ^2 Critical = 14.68



- ❖ $\alpha = 0.10$ Two Tail
- ❖ $Df = 9$
- ❖ χ^2 Critical = 3.325 and 16.92

HYPOTHESIS TESTS



TWO-SAMPLE Z TEST

- ❖ A Two-Sample Z-Test is used to compare the means of two independent samples to determine if there is a significant difference between them.
- ❖ It is used when the population variances (σ_1^2 and σ_2^2) are known OR the sample sizes are large ($n_1 \geq 30$ and $n_2 \geq 30$).

TWO-SAMPLE Z TEST - ASSUMPTIONS

- ❖ Independent Samples:

- ❖ The two samples being compared must be independent of each other.

- ❖ Normal Distribution:

- ❖ The population from which the samples are drawn should follow a normal distribution, or the sample sizes should be large enough ($n \geq 30$) for the Central Limit Theorem to apply.

- ❖ Known Population Variances (σ_1^2 and σ_2^2):

- ❖ The population variances for both groups should be known, or approximated when the sample size is large.

- ❖ Random Sampling:

- ❖ The samples should be randomly selected from their respective populations.

TWO-SAMPLE Z TEST - STEPS

- ❖ 1. State the Hypotheses:
 - ❖ Null Hypothesis (H_0): $\mu_1 = \mu_2$ (The means of both populations are equal).
 - ❖ Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$ (two-tailed test), $\mu_1 > \mu_2$, or $\mu_1 < \mu_2$ (one-tailed test).
- ❖ 2. Set the Significance Level (α):
 - ❖ Typically set at 0.05 for a 95% confidence level.
- ❖ 3. Calculate the Test Statistic (Z):
 - ❖ Use the formula for the z-statistic (given in the next slide).
- ❖ 4. Find the Critical Value:
 - ❖ Use the Z-table to find the critical value based on the significance level and whether the test is one-tailed or two-tailed.
 - ❖ **5. Decision:** Compare the calculated z-statistic with the critical z-value. If the absolute value of the z-statistic exceeds the critical value, reject H_0 .

TWO-SAMPLE Z TEST - STEPS

❖ Z-Statistic Formula:

$$Z = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- ❖ $\bar{x}_1$ and $\bar{x}_2$: Sample means of the two groups.
- ❖ σ_1^2 and σ_2^2 : Population variances of the two groups.
- ❖ n_1 and n_2 : Sample sizes of the two groups.

TWO-SAMPLE Z TEST - EXAMPLE

$$Z = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- ❖ A company wants to compare the average sales performance of two regions. Region 1 has a population mean sales of $\bar{x}_1 = 200$, and Region 2 has $\bar{x}_2 = 190$. The standard deviations for both regions are known to be $\sigma_1 = 15$ and $\sigma_2 = 10$, respectively. A sample of 50 sales records is taken from each region. Test at a 95% confidence level whether the **mean sales are significantly different between the two regions**.
- ❖ Step-by-Step Solution:
- ❖ 1. Hypotheses:
- ❖ $H_0: \mu_1 = \mu_2$ (The mean sales of both regions are equal).
- ❖ $H_1: \mu_1 \neq \mu_2$ (The mean sales of both regions are different).

TWO-SAMPLE Z TEST - EXAMPLE

❖ 2. Significance Level:

- ❖ $\alpha=0.05$ (two-tailed test).

❖ 3. Test Statistic:

$$\text{❖ } Z = \frac{200-190}{\sqrt{\frac{15^2}{50} + \frac{10^2}{50}}} = \frac{10}{\sqrt{4.5+2}} = 3.92$$

❖ 4. Critical Z-Value:

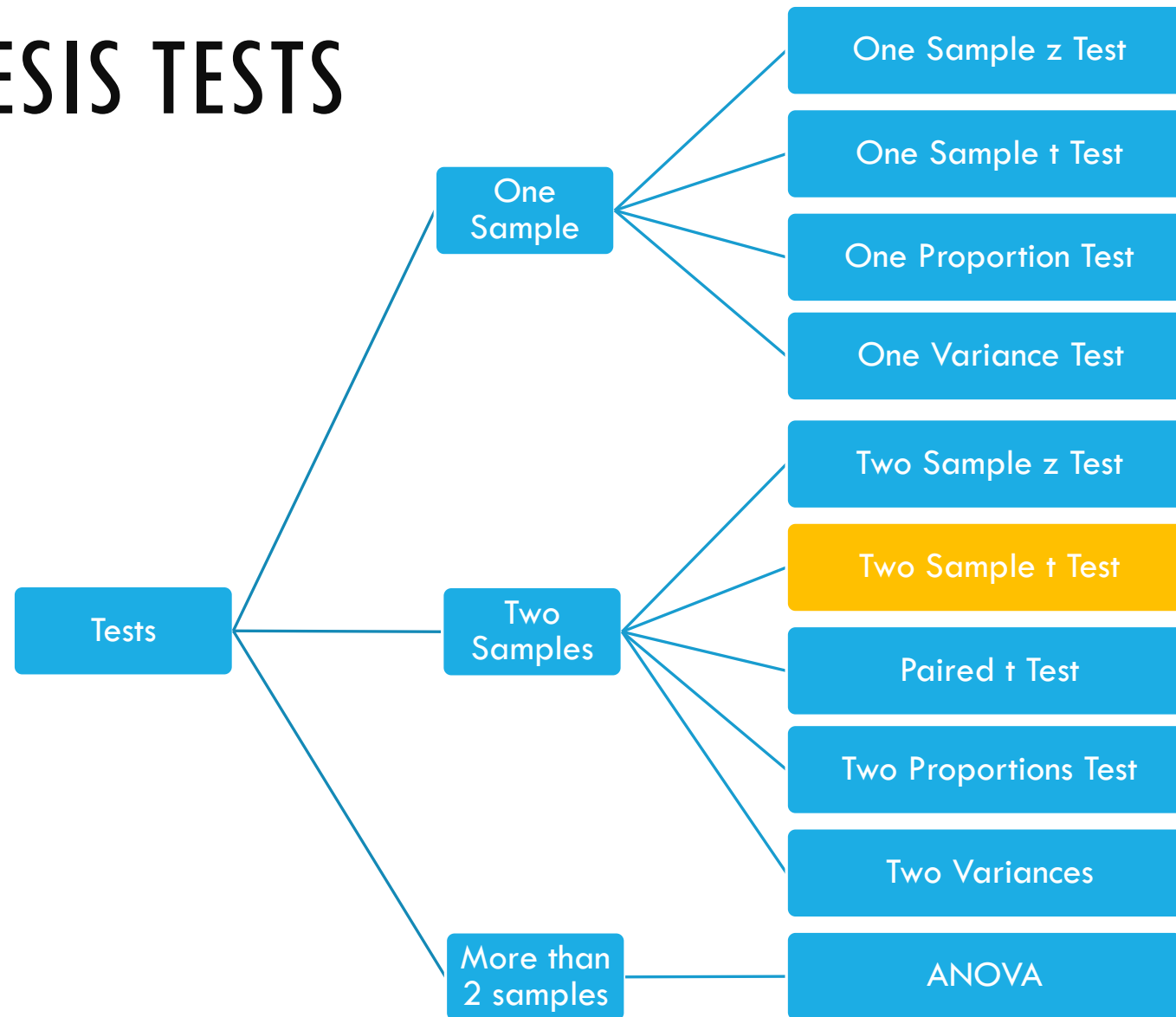
- ❖ For a two-tailed test at $\alpha=0.05$, the critical z-value is ± 1.96 .

❖ 5. Decision:

- ❖ Since $|Z|=3.92$ is greater than 1.96, we reject the null hypothesis.
- ❖ There is enough statistical evidence to suggest that the mean sales in the two regions are significantly different.

$$Z = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

HYPOTHESIS TESTS



TWO-SAMPLE T TEST

- ❖ A Two-Sample t-Test is used to determine whether the means of two independent samples are significantly different from each other.
- ❖ This test is applicable when the population standard deviations are unknown and the samples are small. (works for the large sample size as well)
- ❖ Types of Two-Sample t-Tests:
 - ❖ **Equal Variance (Pooled t-Test):** Assumes that the population variances of both groups are equal.
 - ❖ **Unequal Variance (Welch's t-Test):** Does not assume equal variances.
 - ❖ **Paired t-Test:** Compares means from two related groups (e.g., before and after measurements from the same subjects).

TWO-SAMPLE T TEST (INDEPENDENT)

- ❖ Independent Samples:

- ❖ The two samples being compared must be independent of each other.

- ❖ Normal Distribution:

- ❖ The populations from which the samples are drawn should follow a normal distribution, especially for small sample sizes ($n < 30$).

- ❖ Unknown Population Standard Deviations:

- ❖ The population variances (σ_1^2 and σ_2^2) are unknown and will be estimated from the sample data.

- ❖ Equal or Unequal Variances: Two versions of the two-sample t-test:

- ❖ Equal Variance (Pooled t-Test): Assumes equal variances.
- ❖ Unequal Variance (Welch's t-Test): Does not assume equal variances.

- ❖ Random Sampling:

- ❖ The samples should be randomly selected from their respective populations.

TWO-SAMPLE T TEST - STEPS

❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): $\mu_1 = \mu_2$ (The means of both populations are equal).
- ❖ Alternative Hypothesis (H_1): $\mu_1 \neq \mu_2$ (two-tailed test), $\mu_1 > \mu_2$, or $\mu_1 < \mu_2$ (one-tailed tests).

❖ Set the Significance Level (α):

- ❖ Typically set at 0.05 for a 95% confidence level.

❖ Check the Variance Assumption: (Two Variance Test)

- ❖ If the sample variances are similar, use the equal variance t-test.
- ❖ If the sample variances are significantly different, use the unequal variance t-test (Welch's t-test).

❖ Calculate the Test Statistic (t) / Find the Critical t-Value / Decision:

- ❖ Compare the calculated t-statistic with the critical t-value. If the absolute value of the t-statistic exceeds the critical value, reject H_0 .

TWO-SAMPLE T TEST - STEPS

❖ Is variance for two samples equal?

Yes

$$t_{cal} = \frac{(\bar{x}_1 - \bar{x}_2)}{s_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$df = n_1 + n_2 - 2$$

No

$$t_{cal} = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

$$df = \frac{\left[\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right]^2}{\frac{(s_1^2/n_1)^2}{(n_1 - 1)} + \frac{(s_2^2/n_2)^2}{(n_2 - 1)}}$$

(round down to nearest integer)

TWO-SAMPLE T TEST - EXAMPLE

- ❖ A study is conducted to compare the average test scores of two different teaching methods. A random sample of 16 students from Method A has a mean score of 75 with a variance of 16, and a random sample of 16 students from Method B has a mean score of 80 with a variance of 25. Test at a 95% confidence level whether there is a significant difference in the mean scores of the two methods. (assume equal variance)
- ❖ Step-by-Step Solution:
- ❖ Hypotheses:
 - ❖ $H_0: \mu_1 = \mu_2$ (The mean scores of both methods are equal).
 - ❖ $H_1: \mu_1 \neq \mu_2$ (The mean scores of both methods are different).
- ❖ Significance Level: $\alpha = 0.05$ (two-tailed test).

TWO-SAMPLE T TEST - EXAMPLE

❖ Pooled Variance:

$$S_p^2 = \frac{(16-1) \times 16 + (16-1) \times 25}{16+16-2} = \frac{240+375}{30} = 20.5$$

❖ t-Statistic:

$$t = \frac{75-80}{\sqrt{20.5} \cdot \sqrt{\frac{1}{16} + \frac{1}{16}}} = -\frac{5}{1.601} = -3.12$$

❖ Degrees of Freedom:

$$\text{df} = n_1 + n_2 - 2 = 16 + 16 - 2 = 30$$

$$S_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$t_{cal} = \frac{(\bar{x}_1 - \bar{x}_2)}{s_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

TWO-SAMPLE T TEST - EXAMPLE

❖ t-Statistic:

$$❖ t = \frac{75-80}{\sqrt{20.5} \cdot \sqrt{\frac{1}{16} + \frac{1}{16}}} = -\frac{5}{1.601} = -3.12$$

❖ Degrees of Freedom: $df = n_1 + n_2 - 2 = 20 + 22 - 2 = 40$

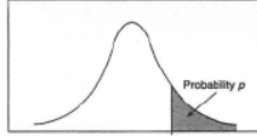
❖ Critical t-Value:

❖ From the t-table, for $df=30$ and $\alpha=0.05$ (two-tailed), the critical value is ± 2.042 .

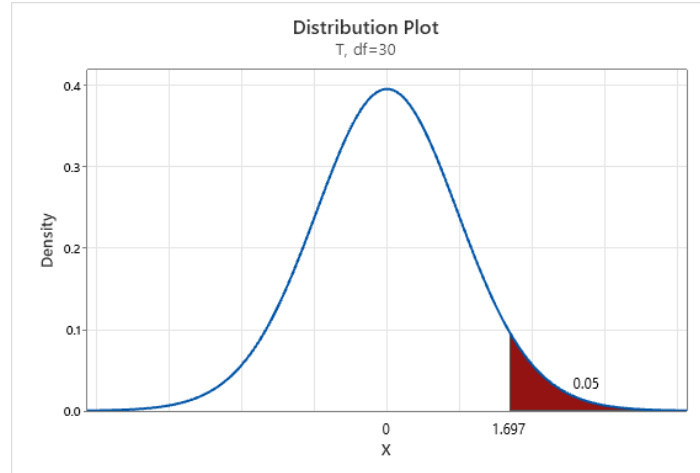
❖ Decision:

- ❖ Since $|t|=3.12$ is greater than the critical value of 2.042, we reject the null hypothesis.
- ❖ There is enough statistical evidence to suggest that the mean scores of the two teaching methods are significantly different.

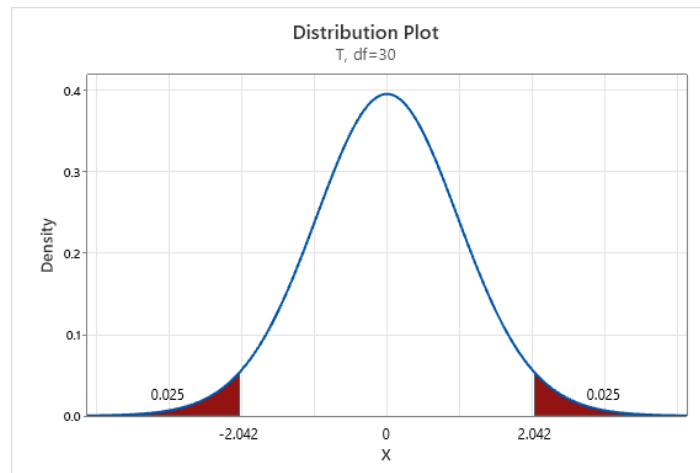
CRITICAL TEST STATISTIC (T VALUE)



df	TAIL PROBABILITY P											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646

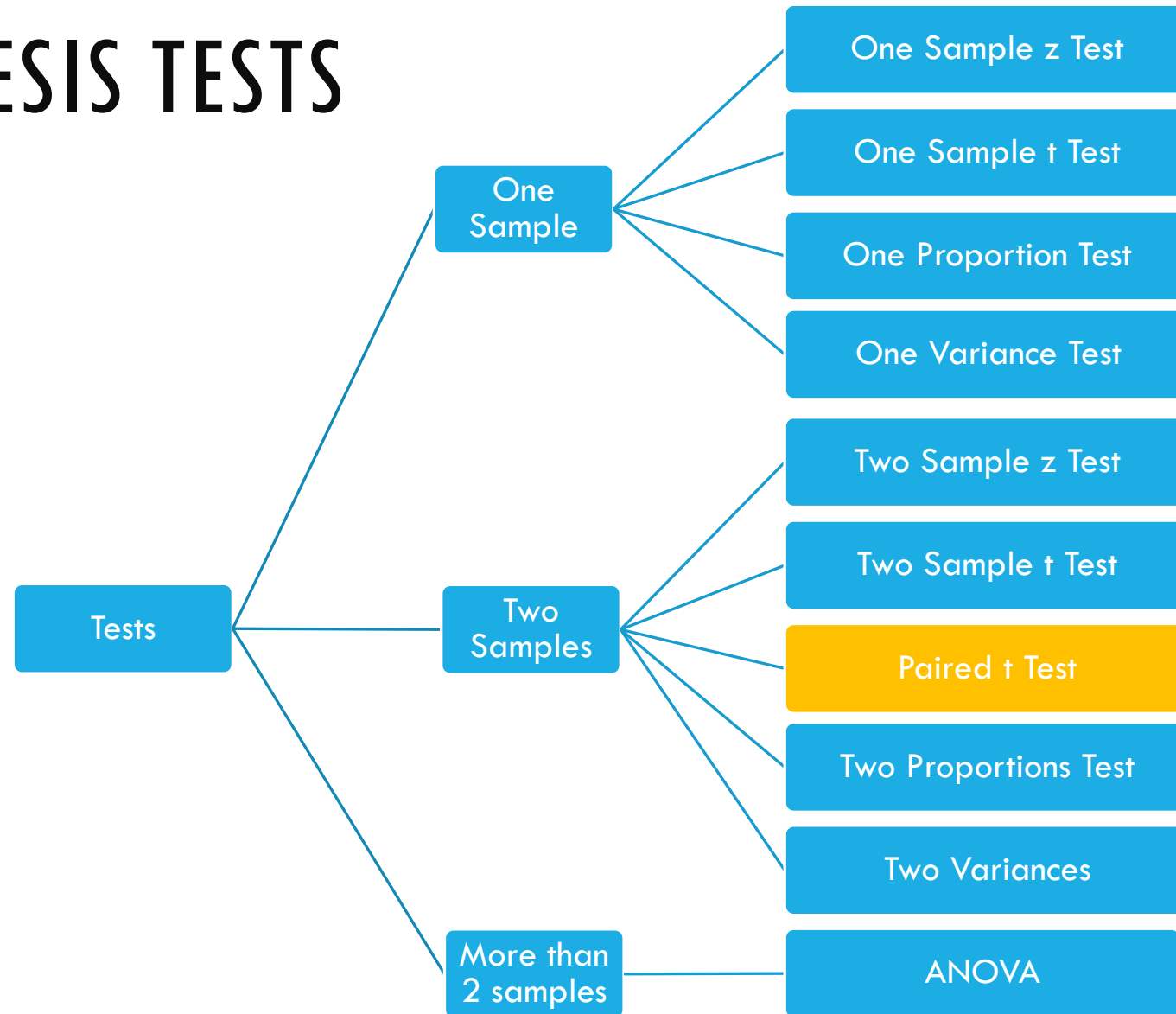


- ❖ $\alpha = 0.05$ One Tails
- ❖ $Df = 30$
- ❖ $t \text{ Critical} = 1.697$



- ❖ $\alpha = 0.05$ Two Tails
- ❖ $Df = 30$
- ❖ $t \text{ Critical} = 2.042$

HYPOTHESIS TESTS



PAIRED T TEST

- ❖ A Paired t-Test is used to compare the means of two related (dependent) groups.
- ❖ This test is typically applied when you want to measure the same subjects before and after a treatment, or when the subjects in one sample are matched with subjects in another sample.
- ❖ Difference Between Independent and Dependent Data:
 - ❖ Independent Data: The observations in one sample are not related to the observations in the other sample (e.g., comparing test scores of students from two different schools).
 - ❖ Dependent Data: The observations in the two groups are related or paired (e.g., measuring a person's weight before and after a diet).

PAIRED T TEST

❖ When to Use:

- ❖ When you have matched pairs of data.
- ❖ When measuring the same group at two different times (e.g., before and after an intervention).
- ❖ When the differences between paired observations are of interest, rather than the individual observations themselves.

❖ Examples:

- ❖ Measuring the blood pressure of patients before and after treatment.
- ❖ Comparing test scores of students before and after taking a course.
- ❖ Evaluating the effect of a new teaching method on the same group of students by comparing their scores before and after the method is introduced.

PAIRED T TEST - STEPS

❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): The mean difference between the paired observations is zero ($\mu_d=0$).
- ❖ Alternative Hypothesis (H_1): The mean difference between the paired observations is not zero ($\mu_d \neq 0$), or directional ($\mu_d > 0$ or $\mu_d < 0$).

❖ Set the Significance Level (α):

- ❖ Typically set at 0.05 for a 95% confidence level.

❖ Test:

- ❖ Compute the difference between the paired observations for each pair: $d_i = X_{\text{after}} - X_{\text{before}}$.
- ❖ Calculate the Test Statistic (t): Use the formula for the paired t-statistic (shown on next slide)
- ❖ Find the Critical t-Value based / Compare the calculated t-statistic with the critical t-value. If the absolute value of the t-statistic exceeds the critical value, reject H_0 .

PAIRED T TEST - FORMULA

❖ Paired t-Test Formula:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

❖ Where:

- ❖ $\bar{d}$: Mean of the differences.
 - ❖ s_d : Standard deviation of the differences.
 - ❖ n : Number of pairs.
- ❖ Degrees of Freedom: $df = n - 1$

PAIRED T TEST - EXAMPLE

- ❖ A teacher wants to evaluate the effectiveness of a new teaching method. She tests the same 10 students before and after the new method is introduced. The test scores before and after the method are as follows:



Student	Before	After
1	70	75
2	80	82
3	90	89
4	85	88
5	78	82
6	88	90
7	92	94
8	84	87
9	86	81
10	89	93

- ❖ Test at a 95% confidence level whether the new teaching method has improved student scores.

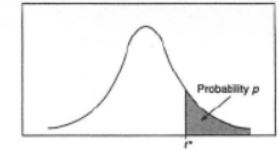
PAIRED T TEST - EXAMPLE

- ❖ Hypotheses:
- ❖ H_0 : The mean difference between test scores is zero ($\mu_d=0$).
- ❖ H_1 : The mean difference between test scores is greater than zero ($\mu_d>0$).
- ❖ Calculate the Differences:
- ❖ $d_i = X_{\text{after}} - X_{\text{before}}$ for each student:

Student	Before	After	Difference (d)
1	70	75	5
2	80	82	2
3	90	89	-1
4	85	88	3
5	78	82	4
6	88	90	2
7	92	94	2
8	84	87	3
9	86	81	-5
10	89	93	4

Sum = 19
Average = 1.9
Sd. Dev = 2.92

PAIRED T TEST - EXAMPLE



Test Statistic: $t = \frac{\bar{d}}{s_d/\sqrt{n}}$

$$\diamond t = \frac{1.9}{\frac{2.92}{\sqrt{10}}} = \frac{1.9}{0.923} = 2.058$$

Degrees of Freedom:

$$\diamond df = n - 1 = 10 - 1 = 9$$

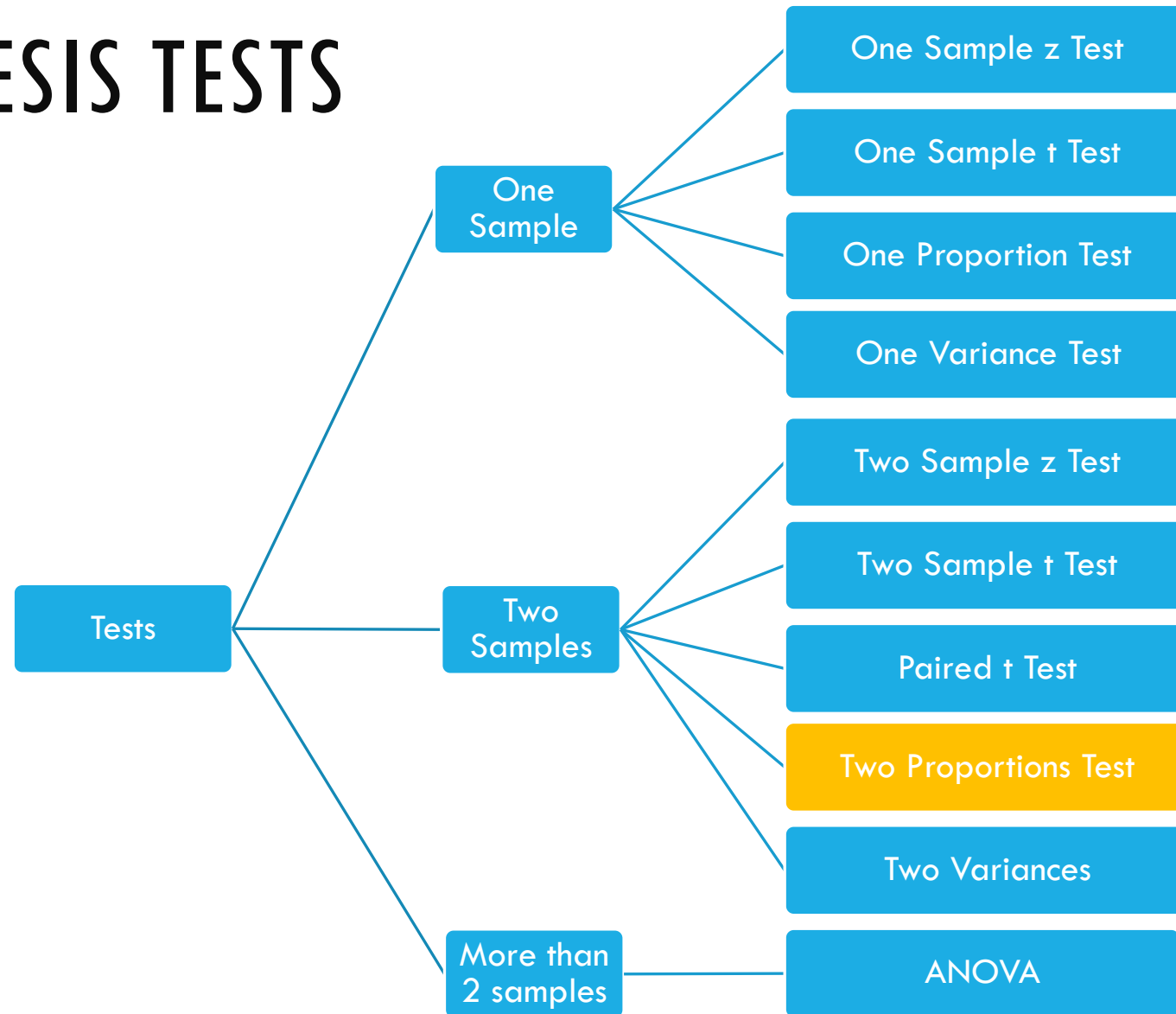
Critical t-Value:

$\diamond$ From the t-table, for $df=9$ and $\alpha=0.05$ (one-tailed), the critical value is 1.833.

Decision: Since $t=2.058$ is greater than the critical value 1.833. we reject the null hypothesis. There is sufficient evidence to conclude that the new teaching method has significantly improved the students' test scores.

df	TAIL PROBABILITY P											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.385	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781

HYPOTHESIS TESTS



TWO PROPORTIONS TEST

- ❖ A Two Proportions Test is used to determine whether there is a significant difference between the proportions of successes in two independent samples.
- ❖ It is often used when you want to compare two groups based on a binary outcome (e.g., success/failure, yes/no).
- ❖ Purpose:
 - ❖ To test if the proportion of successes in one group is different from the proportion in another group.
- ❖ Examples:
 - ❖ Comparing the proportion of patients recovering in two different hospitals.

TWO PROPORTIONS TEST - ASSUMPTIONS

- ❖ Independent Samples:

- ❖ The two samples must be independent of each other.

- ❖ Binary Data:

- ❖ The data should be categorical with two possible outcomes (e.g., success/failure, yes/no).

- ❖ Random Sampling:

- ❖ The samples must be randomly selected from their respective populations.

- ❖ Sample Size:

- ❖ Both samples should be large enough to use the normal approximation to the binomial distribution.
- ❖ This is generally satisfied if $n_1 \cdot p_1 \geq 5$ and $n_1 \cdot (1 - p_1) \geq 5$ and similarly, for n_2 .

TWO PROPORTIONS TEST - STEPS

❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): $p_1 = p_2$ (The proportions are equal).
- ❖ Alternative Hypothesis (H_1): $p_1 \neq p_2$ (two-tailed test), $p_1 > p_2$ or $p_1 < p_2$ (one-tailed tests).

❖ Choose the Significance Level (α):

- ❖ Typically set at 0.05 for a 95% confidence level.

❖ Calculate the Test Statistic (Z):

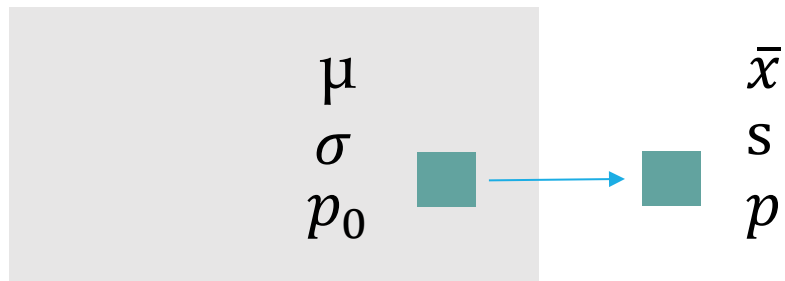
- ❖ Use the formula for the z-statistic (given in the next slide).

❖ Find the Critical Value:

- ❖ Use the Z-table to find the critical value based on the significance level and whether the test is one-tailed or two-tailed.
- ❖ **Decision:** Compare the calculated z-statistic with the critical z-value. If the absolute value of the z-statistic exceeds the critical value, reject H_0 .

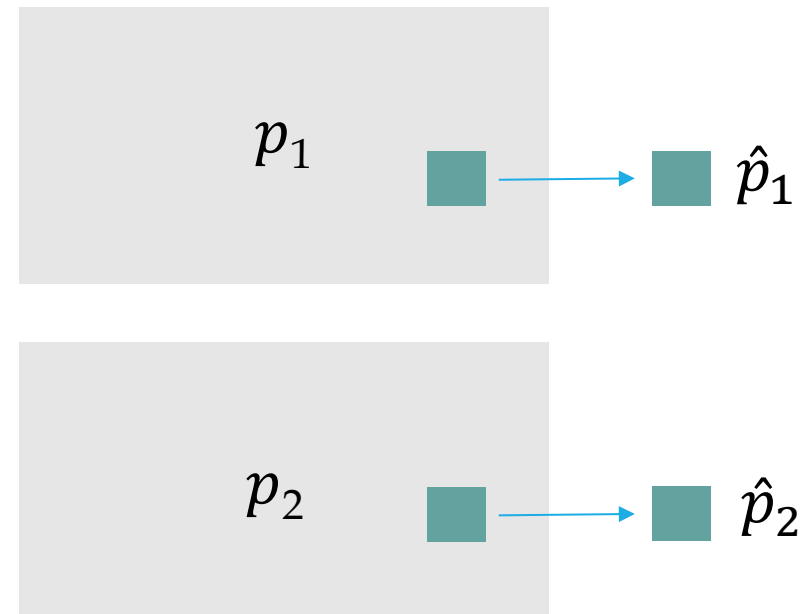
TWO PROPORTIONS TEST - FORMULA

How do we represent sample and population proportions?



One Proportion Test

$$z = \frac{p - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$



$$z = \frac{p - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

TWO PROPORTIONS TEST - FORMULA

Test for no difference between proportions

$$H_0: p_1 - p_2 = 0$$

$$H_a: p_1 - p_2 \neq 0$$

Yes

Pooled

(No difference)

No

Un-pooled

(Specific difference)

$$\bar{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

$$z_{cal} = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\bar{p}(1 - \bar{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$z_{cal} = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}}$$

TWO PROPORTIONS TEST - EXAMPLE

- ❖ A researcher wants to compare the effectiveness of two different drugs. In a clinical trial, Drug A was given to 200 patients, and 120 of them recovered. Drug B was given to 180 patients, and 90 of them recovered. Test at a 95% confidence level whether there is a significant difference in the recovery rates between the two drugs.
- ❖ Step-by-Step Solution:
- ❖ Hypotheses:
 - ❖ $H_0: p_1 = p_2$ (The recovery rates are equal).
 - ❖ $H_1: p_1 \neq p_2$ (The recovery rates are different).
- ❖ Significance Level:
 - ❖ $\alpha = 0.05$ (two-tailed test).

TWO PROPORTIONS TEST - EXAMPLE

❖ Calculate the Sample Proportions:

❖ $\hat{p}_1 = \frac{120}{200} = 0.60$ (Drug A recovery rate).

❖ $\hat{p}_2 = \frac{90}{180} = 0.50$ (Drug B recovery rate).

❖ Pooled Proportion:

❖ $\hat{p} = \frac{120+90}{200+180} = \frac{210}{380} = 0.5526$

❖ Z-Statistic:

❖
$$Z = \frac{0.60 - 0.50}{\sqrt{0.5526 \times (1 - 0.5526) \times \left(\frac{1}{200} + \frac{1}{180}\right)}} = \frac{0.10}{\sqrt{0.5526 \times 0.4474 \times 0.01055}} = \frac{0.10}{0.0511} = 1.957$$

$$\bar{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{(n_1 + n_2)}$$

$$Z_{cal} = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\bar{p}(1 - \bar{p}) \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

TWO PROPORTIONS TEST - EXAMPLE

❖ Z-Statistic:

$$\text{❖ } Z = \frac{0.60 - 0.50}{\sqrt{0.5526 \times (1 - 0.5526) \times \left(\frac{1}{200} + \frac{1}{180}\right)}} = \frac{0.10}{\sqrt{0.5526 \times 0.4474 \times 0.01055}} = \frac{0.10}{0.0511} = 1.957$$

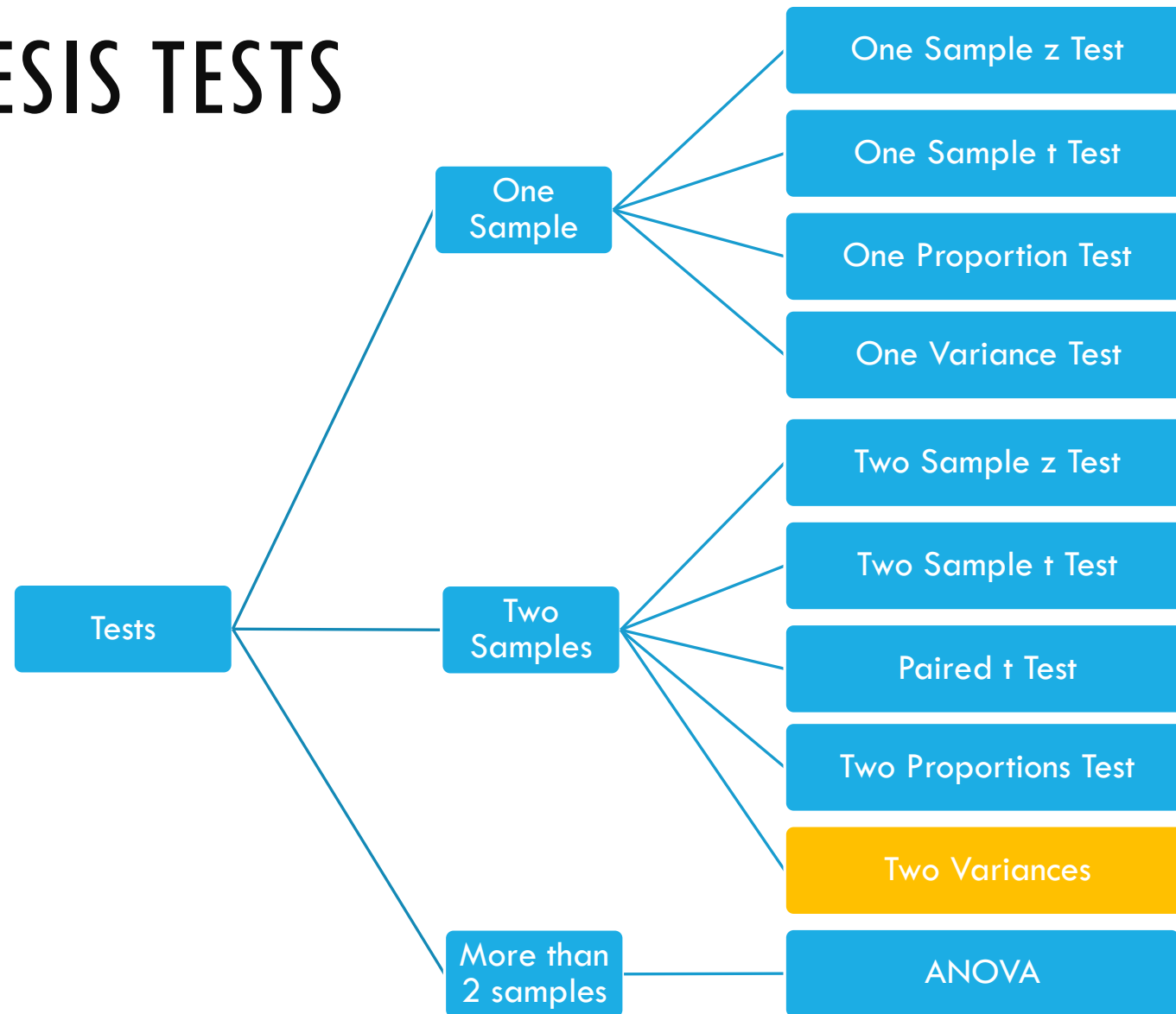
❖ Critical Z-Value:

❖ For a two-tailed test at $\alpha=0.05$, the critical z-value is ± 1.96 .

❖ Decision:

❖ Since $|Z|=1.957$ is less than the critical value of 1.96, we fail to reject the null hypothesis. There is not enough statistical evidence to conclude that the recovery rates for the two drugs are significantly different.

HYPOTHESIS TESTS



TWO VARIANCE TEST

- ❖ A Two Variance Test is used to determine whether the variances of two independent populations are significantly different from each other.
- ❖ The F-Test is typically used for this comparison, where the test statistic follows an F-distribution.
- ❖ Purpose:
 - ❖ To test whether the variability (spread) of two different populations is the same or significantly different.
- ❖ Examples:
 - ❖ Comparing variability in product weights from two manufacturing plants.
 - ❖ Assessing whether two teaching methods result in different variability in student performance.

TWO VARIANCE TEST - ASSUMPTIONS

- ❖ Independent Samples:

- ❖ The two samples being compared must be independent of each other.

- ❖ Normal Distribution:

- ❖ The populations from which the samples are drawn should follow a normal distribution.

- ❖ Random Sampling:

- ❖ The samples must be randomly selected from their respective populations.

- ❖ Two-Tailed or One-Tailed Test:

- ❖ A two-tailed test is used when we want to check whether the variances are different.
 - ❖ A one-tailed test is used if we want to test if one variance is larger or smaller than the other.

TWO VARIANCE TEST - STEPS

❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$ (The variances of the two populations are equal).
- ❖ Alternative Hypothesis (H_1): $\sigma_1^2 \neq \sigma_2^2$ (two-tailed test), $\sigma_1^2 > \sigma_2^2$, or $\sigma_1^2 < \sigma_2^2$ (one-tailed tests).

❖ Set the Significance Level (α):

- ❖ Typically set at 0.05 for a 95% confidence level.

❖ Calculate the Test Statistic (F):

- ❖ Use the formula for the F-ratio (shown in the next slide).

❖ Find the Critical F-Value:

- ❖ Use the F-distribution table to find the critical value based on the significance level and the degrees of freedom for each sample ($df1 = n_1 - 1$ and $df2 = n_2 - 1$).
- ❖ **Decision:** Compare the calculated F-statistic with the critical F-value. If the calculated F-statistic exceeds the critical value, reject H_0 .

TWO VARIANCE TEST - FORMULA

❖ F-Test Statistic Formula:

$$❖ F = \frac{s_1^2}{s_2^2}$$

❖ Where:

❖ s_1^2 : Sample variance of the first group (typically the larger variance is placed in the numerator).

❖ s_2^2 : Sample variance of the second group.

❖ Degrees of Freedom:

❖ $df1 = n_1 - 1$ (for the numerator) / $df2 = n_2 - 1$ (for the denominator).

❖ F-Distribution: The F-test statistic follows an F-distribution with degrees of freedom $df1$ and $df2$.

TWO VARIANCE TEST - EXAMPLE

- ❖ A manufacturer wants to compare the variability in the weights of products from two different production lines. A sample of 10 products from Line 1 has a variance of 4.5 grams², and a sample of 12 products from Line 2 has a variance of 2 grams². Test at a 90% confidence level whether the variances are different.
- ❖ Step-by-Step Solution:
- ❖ Hypotheses:
- ❖ $H_0: \sigma_1^2 = \sigma_2^2$ (The variances are equal).
- ❖ $H_1: \sigma_1^2 \neq \sigma_2^2$ (The variances are different).
- ❖ Significance Level: $\alpha = 0.10$ (two-tailed test).

TWO VARIANCE TEST - EXAMPLE

❖ Calculate the F-Statistic:

$$❖ F = \frac{s_1^2}{s_2^2} = \frac{4.5}{2} = 2.25$$

❖ Degrees of Freedom:

$$❖ df1 = n_1 - 1 = 10 - 1 = 9 \quad / \quad df2 = n_2 - 1 = 12 - 1 = 11$$

❖ Critical F-Value:

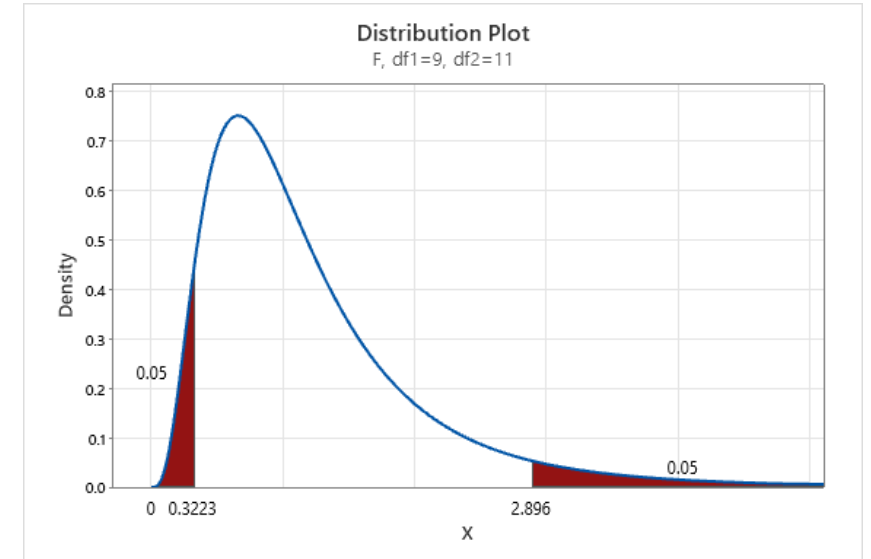
❖ From the F-table, for $df1=9$ and $df2=11$, and $\alpha=0.05$ (two-tailed), the critical values are:

❖ Upper critical value: $F_{(0.05,9,11)} = 2.90$.

❖ Lower critical value: $F_{(0.95,9,11)} = 1 / F_{(0.05,11,9)} = 1 / 3.105 = 0.332$

❖ Since $0.332 < F=2.25 < 2.90$ we fail to reject the null hypothesis.

❖ There is not enough statistical evidence to conclude that the variances are significantly different.



TWO VARIANCE TEST - EXAMPLE

❖ Critical F-Value:

- ❖ From the F-table, for $df1=9$ and $df2=11$, and $\alpha=0.05$ (two-tailed), the critical values are:

❖ Upper critical value: $F_{(0.05,9,11)} = 2.90$.

❖ Lower critical value: $F_{(0.95,9,11)} = 1 / F_{(0.05,11,9)} = 1 / 3.105 = 0.332$

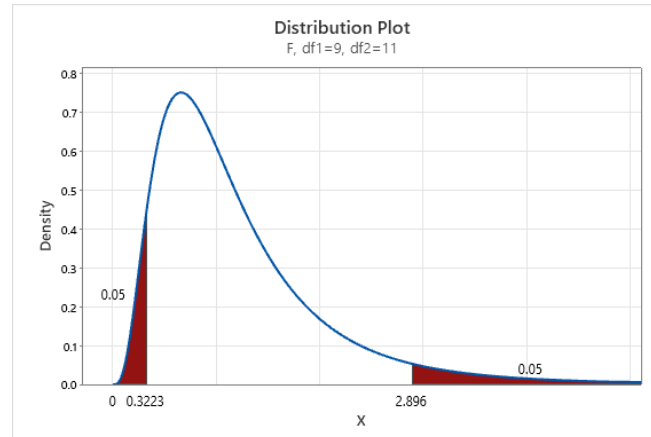
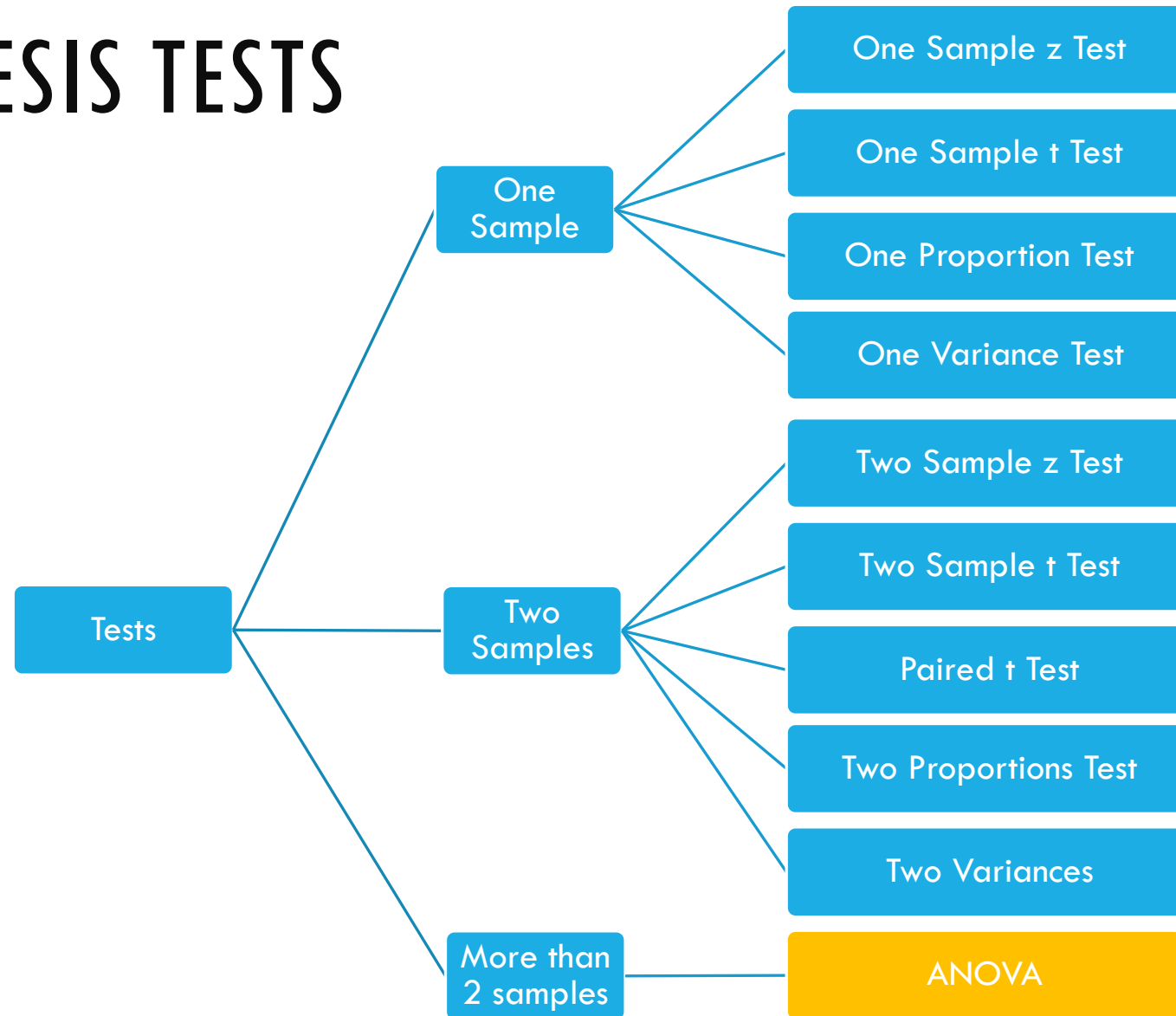


Table 6: F-table of Critical Values of $\alpha = 0.05$ for $F(df1, df2)$



	DF1=1	2	3	4	5	6	7	8	9	10	12	15	20
DF2=1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57

HYPOTHESIS TESTS



ANOVA

- ❖ ANOVA (Analysis of Variance) is a statistical method used to test whether there are any significant differences between the **means of three or more independent groups**.
- ❖ It helps determine if the observed differences between group means are due to random chance or if they reflect actual differences in the populations.
- ❖ Purpose:
 - ❖ To assess whether the means of several groups are statistically different from one another by comparing their variances.
- ❖ Example: Testing if different teaching methods result in different average student scores.

WHY NOT USE MULTIPLE T-TESTS?

❖ Increased Type I Error:

- ❖ When performing multiple t-tests, the probability of making a Type I error (false positive) increases with each test.
- ❖ For example, if you conduct three independent t-tests with a significance level of 0.05, the overall chance of a Type I error becomes much higher than 5%.
- ❖ Example: Conducting 3 t-tests at $\alpha=0.05$ gives a cumulative error rate of $1 - (0.95)^3 \approx 0.1414$.

❖ Less Efficient:

- ❖ Multiple t-tests require performing several calculations, making the process longer and more complex, especially as the number of groups increases.

❖ Post-hoc Tests:

- ❖ If ANOVA finds a significant difference, you can follow up with post-hoc tests (such as Tukey's test) to identify which groups differ, without inflating the error rate.

ANOVA - ASSUMPTIONS

- ❖ Independent Samples:

- ❖ The groups being compared must be independent of one another.

- ❖ Normal Distribution:

- ❖ The populations from which the samples are drawn should follow a normal distribution, especially for small sample sizes.

- ❖ Equal Variances (Homoscedasticity):

- ❖ The variances across the different groups should be approximately equal (homogeneity of variances).

- ❖ Random Sampling:

- ❖ The samples must be randomly selected from their respective populations.

ANOVA - TYPES

❖ One-Way ANOVA:

- ❖ Used when comparing the means of three or more independent groups based on one factor (e.g., different teaching methods).

❖ Two-Way ANOVA:

- ❖ Used when comparing the means of groups based on two factors (e.g., comparing the effect of teaching methods and gender on student scores).

ANOVA - STEPS

❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): $\mu_1 = \mu_2 = \mu_3 = \dots$ (The means of all groups are equal).
- ❖ Alternative Hypothesis (H_1): **At least one group** mean is different.

❖ Set the Significance Level (α):

- ❖ Typically set at 0.05 for a 95% confidence level.

❖ Calculate the Test Statistic (F):

- ❖ Use the formula for the F-ratio (shown in the next slide).

❖ Find the Critical F-Value:

- ❖ Use the F-distribution table to find the critical value based on the significance level and degrees of freedom.
- ❖ **Decision:** Compare the calculated F-statistic with the critical F-value. If the F-statistic exceeds the critical value, reject H_0 .

ANOVA - FORMULA

❖ F-Statistic Formula:

$$❖ F = \frac{\text{Between-group variance (MSB)}}{\text{Within-group variance (MSW)}}$$

❖ Where:

- ❖ MSB (Mean Square Between): Measures the variance between the groups.
- ❖ MSW (Mean Square Within): Measures the variance within the groups.

ANOVA - FORMULA

- ❖ Calculate the Total Sum of Squares (SST):

- ❖ $SST = \sum_{i=1}^n (X_{ij} - \bar{X})^2$

- ❖ Measures the total variability in the data.

- ❖ Calculate the Sum of Squares Between (SSB):

- ❖ $SSB = \sum_{i=1}^k n_i (\bar{X}_i - \bar{X})^2$

- ❖ Measures the variance due to differences between group means.

$$\bar{X} = 152.44$$

- ❖ Calculate the Sum of Squares Within (SSW):

- ❖ $SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} n_i (\bar{X}_{ij} - \bar{X}_i)^2$

- ❖ Measures the variance within each group.

Machine 1	Machine 2	Machine 3
150	153	156
151	152	154
152	148	155
152	151	156
151	149	157
150	152	155
$\bar{x}_1 = 151.00$	$\bar{x}_2 = 150.83$	$\bar{x}_3 = 155.50$

ANOVA - FORMULA

- ❖ Calculate the Mean Square Between (MSB):

- ❖ $MSB = \frac{SSB}{k-1}$

- ❖ Calculate the Mean Square Within (MSW):

- ❖ $MSW = \frac{SSW}{n-k}$

- ❖ Calculate the F-Statistic:

- ❖ $F = \frac{MSB}{MSW}$

- ❖ k is the number of groups.

- ❖ n is the total number of observations.

ANOVA - EXAMPLE

- ❖ Problem Statement: A manufacturer wants to compare the average weights of products produced by three different machines.
- ❖ The weights from each machine are recorded in the table below:

Machine 1	Machine 2	Machine 3
150	153	156
151	152	154
152	148	155
152	151	156
151	149	157
150	152	155
Mean1= 151	Mean2=150.83	Mean3=155.50

$$\bar{X} = 152.44$$

ANOVA - EXAMPLE

❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): The mean weights of the products produced by all three machines are equal.
- ❖ $H_0: \mu_1 = \mu_2 = \mu_3$
- ❖ Alternative Hypothesis (H_1): **At least one of the machines** produces products with a different mean weight.

❖ Calculate the Group Means:

- ❖ Machine 1 Mean: $\overline{X}_1 = 151$
- ❖ Machine 2 Mean: $\overline{X}_2 = 150.83$
- ❖ Machine 3 Mean: $\overline{X}_3 = 155.50$

❖ Grand Mean:

- ❖ $\overline{X} = (150 + 151 + 152 + \dots + 155 + 156 + 157 + 155) / 18 = 152.44$

ANOVA - EXAMPLE

- ❖ Calculate the Sum of Squares Between (SSB):
 - ❖ SSB measures the variance between the group means and the grand mean:
 - ❖ $SSB = 6 \times ((151 - 152.44)^2 + (150.83 - 152.44)^2 + (155.50 - 152.44)^2)$
 - ❖ $SSB = 6 \times (2.07 + 2.59 + 9.36) = 6 \times 14.02 = 84.12$
- ❖ Calculate the Sum of Squares Within (SSW):
 - ❖ SSW measures the variance within each group:
 - ❖ $SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} n_i (\bar{X}_{ij} - \bar{X}_i)^2$
 - ❖ $SSW = 22.33$. (See the next slide for this calculation)

ANOVA - EXAMPLE

Machine 1	$x_1 - \bar{x}_1$	$Sqr(x_1 - \bar{x}_1)$	Machine 2	$x_2 - \bar{x}_2$	$Sqr(x_2 - \bar{x}_2)$	Machine 3	$x_3 - \bar{x}_3$	$Sqr(x_3 - \bar{x}_3)$	
150.00	-1.00	1.00	153.00	2.17	4.69	156.00	0.50	0.25	
151.00	0.00	0.00	152.00	1.17	1.36	154.00	-1.50	2.25	
152.00	1.00	1.00	148.00	-2.83	8.03	155.00	-0.50	0.25	
152.00	1.00	1.00	151.00	0.17	0.03	156.00	0.50	0.25	
151.00	0.00	0.00	149.00	-1.83	3.36	157.00	1.50	2.25	
150.00	-1.00	1.00	152.00	1.17	1.36	155.00	-0.50	0.25	
151.00			150.83			155.50			152.44
		4.00			18.83			5.50	

$$\diamond SS_{\text{within}} = 4.00 + 18.83 + 5.50 = 28.33$$

ANOVA - EXAMPLE

❖ Mean Square Between (MSB):

$$❖ MSB = \frac{SSB}{k-1} = \frac{84.12}{3-1} = 42.06$$

❖ Mean Square Within (MSW):

$$❖ MSW = \frac{SSW}{n-k} = \frac{28.33}{18-3} = \frac{28.33}{15} = 1.89$$

❖ F-Statistic:

$$❖ F = \frac{MSB}{MSW} = \frac{42.06}{1.89} = 22.25$$

ANOVA - EXAMPLE

Table 6: F-table of Critical Values of $\alpha = 0.05$ for $F(df1, df2)$



❖ Critical F-Value:

- ❖ From the F-distribution table, for $df1=2$ and $df2=15$, and $\alpha=0.05$, the critical F-value is 3.68.
- ❖ Since the calculated F-value 22.25 is much greater than the critical value 3.68, we reject the null hypothesis.

❖ Conclusion:

- ❖ There is a significant difference in the mean weights of products produced by the three machines. At least one machine produces products with a different mean weight.

	DF1=1	2	3	4	5	6	7	8	9	10	12	15	20
DF2=1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57

ANOVA – EXAM PROBLEM

- ❖ A company uses ANOVA to compare the output from three machines (Machine A, Machine B, and Machine C). The partially filled ANOVA summary table is provided below. Use the information given in the table to fill in the missing values.

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Statistic
Between Groups	80	?	?	?
Within Groups	?	?	3	
Total	?	17		

ANOVA – EXAM PROBLEM

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Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Statistic
Between Groups	80	? 2	?	?
Within Groups	?	? 15	3	
Total	?	17		

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Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Statistic
Between Groups	80	? 2	? 40	? 13.33
Within Groups	?	? 15	3	
Total	?	17		

ANOVA – EXAM PROBLEM

- ❖ A company uses ANOVA to compare the output from three machines (Machine A, Machine B, and Machine C). The partially filled ANOVA summary table is provided below. Use the information given in the table to fill in the missing values.

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Statistic
Between Groups	80	? 2	? 40	? 13.33
Within Groups	? 45	? 15	3	
Total	? 125	17		

GOODNESS-OF-FIT TEST

- ❖ A Goodness-of-Fit Test is used to determine how well a set of observed data fits a particular theoretical distribution.
- ❖ Purpose:
 - ❖ To test whether the distribution of a categorical variable follows a hypothesized distribution.
- ❖ Examples:
 - ❖ Testing whether a die is fair (i.e., all six sides have equal probability).
 - ❖ Testing if customer preferences for a product follow a specific pattern.
- ❖ Null Hypothesis (H_0):
 - ❖ The observed data follows the expected distribution.
- ❖ Alternative Hypothesis (H_1):
 - ❖ The observed data does not follow the expected distribution.

GOODNESS-OF-FIT TEST — WHEN TO USE?

- ❖ Categorical Data:

- ❖ The data consists of categories (e.g., color preference, types of products purchased).

- ❖ Comparing Observed to Expected Frequencies:

- ❖ You have a theoretical distribution or expected frequencies, and you want to see if the observed frequencies significantly differ.

- ❖ Sample Size:

- ❖ The test is most appropriate when you have a sufficiently large sample (generally, all expected frequencies should be greater than 5).

- ❖ Discrete Distribution:

- ❖ Goodness-of-Fit is typically used for discrete distributions (e.g., testing uniform, binomial, or Poisson distributions)..

GOODNESS-OF-FIT TEST — STEPS

- ❖ State the Hypotheses:

- ❖ H_0 : The observed data follows the expected distribution.
- ❖ H_1 : The observed data does not follow the expected distribution.

- ❖ Calculate the Expected Frequencies:

- ❖ Based on the theoretical distribution, calculate the expected frequency for each category.

- ❖ Calculate the Chi-Square Statistic and degrees of freedom

- ❖ See the formula on next slide

- ❖ Find the Critical Value:

- ❖ Use the Chi-Square distribution table with the appropriate degrees of freedom and significance level (α , typically 0.05).
- ❖ **Decision:** If the calculated χ^2 statistic is greater than the critical value from the table, reject the null hypothesis.

GOODNESS-OF-FIT TEST — FORMULA

❖ Chi-Square Test Statistic:

$$\chi^2 = \sum \left(\frac{(O_i - E_i)^2}{E_i} \right)$$

❖ Where:

- ❖ O_i = Observed frequency in category i
- ❖ E_i = Expected frequency in category i
- ❖ The sum is taken over all categories.

❖ Degrees of Freedom:

- ❖ $df = k - 1$
- ❖ Where: k = Number of categories.

GOODNESS-OF-FIT TEST — EXAMPLE

❖ A company produces candy in 5 colors. Based on market research, they expect the colour distribution to be uniform (i.e., 20% of candies in each colour). After inspecting a sample of 100 candies, they observe the following frequencies:

❖ Step-by-Step Solution:

❖ Hypotheses:

- ❖ H_0 : The colour distribution of the candies is uniform (20% for each colour).
- ❖ H_1 : The colour distribution of the candies is not uniform.

Color	Observed (O)	Expected (E)
Red	25	20
Blue	15	20
Green	20	20
Yellow	18	20
Orange	22	20

GOODNESS-OF-FIT TEST — EXAMPLE

$$\chi^2 = \sum \left(\frac{(O_i - E_i)^2}{E_i} \right)$$

❖ Calculate the Chi-Square Statistic:

$$\chi^2 = \frac{(25-20)^2}{20} + \frac{(15-20)^2}{20} + \frac{(20-20)^2}{20} + \frac{(18-20)^2}{20} + \frac{(22-20)^2}{20}$$

$$\chi^2 = \frac{25}{20} + \frac{25}{20} + \frac{0}{20} + \frac{4}{20} + \frac{4}{20} = 1.25 + 1.25 + 0 + 0.20 + 0.20 = 2.90$$

❖ Degrees of Freedom:

$$\text{df} = k - 1 = 5 - 1 = 4$$

Color	Observed (O)	Expected (E)
Red	25	20
Blue	15	20
Green	20	20
Yellow	18	20
Orange	22	20

GOODNESS-OF-FIT TEST – EXAMPLE

- ❖ Calculate the Chi-Square Statistic: 2.90 / Degrees of Freedom: 4
- ❖ Critical Value:
 - ❖ For $df=4$ and $\alpha=0.05$, the critical value from the Chi-Square table is 9.49.
 - ❖ Since $\chi^2=2.90$ is less than the critical value 9.49, we fail to reject the null hypothesis.
- ❖ Conclusion:
 - ❖ There is no significant difference between the observed and expected colour distribution of the candies, and the data suggests the distribution is uniform.

Percentage Points of the Chi-Square Distribution									
Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.88	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48

CONTINGENCY TABLES

- ❖ A contingency table is a type of table used in statistics to display the frequency distribution of variables. It shows the relationship between categorical variables.
- ❖ Purpose:
 - ❖ Contingency tables allow us to analyze the association or independence between two categorical variables.
- ❖ Common Use:
 - ❖ Chi-Square Test for Independence is often used with contingency tables to test whether two variables are independent.

CONTINGENCY TABLES

❖ Rows and Columns:

- ❖ The table is organized into rows representing one categorical variable and columns representing another categorical variable.

❖ Cell Frequencies:

- ❖ Each cell of the table contains the frequency (count) of observations that correspond to a specific combination of row and column categories.

	Category 1	Category 2	Category 3	Total
Group A	15	20	10	45
Group B	25	30	20	75
Total	40	50	30	120

CONTINGENCY TABLES - STEPS

❖ State the Hypotheses:

- ❖ Null Hypothesis (H_0): The two categorical variables are independent.
- ❖ Alternative Hypothesis (H_1): The two categorical variables are dependent (associated).

❖ Calculate the Expected Frequencies:

- ❖ The expected frequency for each cell is calculated using the formula:

- ❖
$$E_{ij} = \frac{\text{Row Total}_i \times \text{Column Total}_j}{\text{Grand Total}}$$

❖ Calculate the Chi-Square Statistic:

- ❖
$$\chi^2 = \sum \left(\frac{(O_{ij} - E_{ij})^2}{E_{ij}} \right)$$

- ❖ Where: O_{ij} is the observed frequency in the cell and E_{ij} is the expected frequency for the cell.

CONTINGENCY TABLES - STEPS

- ❖ Degrees of Freedom (df):
- ❖ $df = (r - 1) \times (c - 1)$
- ❖ Where:
 - ❖ r is the number of rows.
 - ❖ c is the number of columns.
- ❖ Decision:
 - ❖ Compare the calculated χ^2 statistic to the critical value from the Chi-Square distribution table. If the calculated value exceeds the critical value, reject H_0 .

CONTINGENCY TABLES - EXAMPLE

- ❖ Problem: A company wants to know if there is a relationship between customer satisfaction (Satisfied/Not Satisfied) and product quality (High/Low). The data from 100 customers is shown in the table below.

	Satisfied	Not Satisfied	Total
High Quality	40	5	45
Low Quality	20	35	55
Total	60	40	100

- ❖ Step-by-Step Solution:
- ❖ Hypotheses:
 - ❖ H_0 : Customer satisfaction is independent of product quality.
 - ❖ H_1 : Customer satisfaction is dependent on product quality.

CONTINGENCY TABLES - EXAMPLE

❖ Calculate Expected Frequencies:

❖ The expected frequencies for each cell are calculated as:

$$❖ E_{ij} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

$$❖ E_{11} = \frac{45 \times 60}{100} = 27$$

$$❖ E_{12} = \frac{45 \times 40}{100} = 18$$

$$❖ E_{21} = \frac{55 \times 60}{100} = 33$$

$$❖ E_{22} = \frac{55 \times 40}{100} = 22$$

	Satisfied	Not Satisfied	Total
High Quality	40	5	45
Low Quality	20	35	55
Total	60	40	100

CONTINGENCY TABLES - EXAMPLE

❖ Calculate Expected Frequencies:

❖ The expected frequencies for each cell are calculated as:

$$\chi^2 = \sum \left(\frac{(O_{ij} - E_{ij})^2}{E_{ij}} \right)$$

❖ $E_{ij} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$

❖ $E_{11} = \frac{45 \times 60}{100} = 27$

❖ $E_{12} = \frac{45 \times 40}{100} = 18$

❖ $E_{21} = \frac{55 \times 60}{100} = 33$

❖ $E_{22} = \frac{55 \times 40}{100} = 22$

OBSERVED	Satisfied	Not Satisfied	Total
High Quality	40	5	45
Low Quality	20	35	55
Total	60	40	100

EXPECTED	Satisfied	Not Satisfied	Total
High Quality	27	18	45
Low Quality	33	22	55
Total	60	40	100

CONTINGENCY TABLES - EXAMPLE

$$\chi^2 = \sum \left(\frac{(O_{ij} - E_{ij})^2}{E_{ij}} \right)$$

❖ Calculate the Chi-Square Statistic:

$$\chi^2 = \frac{(40-27)^2}{27} + \frac{(5-18)^2}{18} + \frac{(20-33)^2}{33} + \frac{(35-22)^2}{22}$$

$$\chi^2 = 6.26 + 9.39 + 5.12 + 7.68 = 28.45$$

❖ Degrees of Freedom:

$$df = (r-1) \times (c-1) = (2-1) \times (2-1) = 1$$

OBSERVED	Satisfied	Not Satisfied	Total
High Quality	40	5	45
Low Quality	20	35	55
Total	60	40	100

EXPECTED	Satisfied	Not Satisfied	Total
High Quality	27	18	45
Low Quality	33	22	55
Total	60	40	100

CONTINGENCY TABLES - EXAMPLE

❖ Calculate the Chi-Square Statistic: $=28.45$

❖ Degrees of Freedom = 1

❖ Critical Value:

❖ From the Chi-Square distribution table, for $df=1$ and $\alpha=0.05$, the critical value is 3.84.

❖ Decision:

❖ Since $\chi^2=28.45$ is much greater than the critical value 3.84, we reject the null hypothesis.

❖ Conclusion:

❖ There is a significant relationship between customer satisfaction and product quality.
Customers are much more likely to be satisfied with high-quality products than low-quality products.

Percentage Points of the Chi-Square Distribution									
Degrees of Freedom	Probability of a larger value of χ^2								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
1	0.000	0.004	0.016	0.102	0.455	1.32	2.71	3.84	6.63
2	0.020	0.103	0.211	0.575	1.386	2.77	4.61	5.99	9.21
3	0.115	0.352	0.584	1.212	2.366	4.11	6.25	7.81	11.34
4	0.297	0.711	1.064	1.923	3.357	5.39	7.78	9.49	13.28
5	0.554	1.145	1.610	2.675	4.351	6.63	9.24	11.07	15.09
6	0.872	1.635	2.204	3.455	5.348	7.84	10.64	12.59	16.81
7	1.239	2.167	2.833	4.255	6.346	9.04	12.02	14.07	18.48